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O'REILLY

1.1 What is Artificial Intelligence?

In 2004, John McCarthy defined artificial intelligence (AI) as the science and engineering of making intelligent machines, especially intelligent computer programs. However, much before this definition was coined, the birth of AI was marked by Alan Turing's seminal work, '*Computing Machinery and Intelligence*,' published in 1950. Alan Turing, also known as the 'father of computer science,' raised questions like 'Can machines think?' in this paper. Continuing with his work, he later proposed the 'Turing Test', where a human interrogator would try to distinguish between a computer and human text response.

Few languages that are popularly used to code AI applications are R, Python and Java.

From a layman's view, artificial intelligence (AI), simply means the intelligence demonstrated by machines that help them to mimic the actions of humans. AI simulates natural intelligence in machines that are programmed to learn from experiences, adjust to new inputs and perform human-like tasks.

From a researcher's view, AI is a set of algorithms that generates results without having to be explicitly instructed to do so, thereby making machines capable of thinking and acting rationally and humanely.

Most AI examples, from chess-playing computers to self-driving cars, heavily depend on deep learning and natural language processing techniques.

AI applications perform specialized tasks by processing large amounts of data and recognizing patterns in them. Besides learning from experience, AI applications can recognize objects, understand and respond to language and make decisions to solve real-world problems. AI perform tasks that normally require human intelligence. The main focus of this technology is to build machines and algorithms which are capable of performing computational tasks that would otherwise require human-like brain functions.

While computers execute certain tasks like sorting, computing, memorizing, indexing, finding patterns, etc. far better than humans, they still cannot beat human skills for identifying emotions, recognizing faces, communication and conversation. This is where AI will play a crucial role to enable machines achieve human capabilities.

NITI Aayog: The National Strategy for Artificial Intelligence has defined AI as follows:

AI refers to the ability of machines to perform cognitive tasks like thinking, perceiving, learning, problem solving and decision-making. Initially conceived as a technology that could mimic human intelligence, AI has evolved in ways that far exceed its original conception. With incredible advances made in data collection, processing and computation power, intelligent systems can now be deployed to take over a variety of tasks, enable connectivity and enhance productivity.

1.1.1 How Does AI Work?

AI systems work effectively when fed with a large amount of labelled training data. This data is thoroughly analysed to discover correlations and patterns. These patterns are then used to make predictions about fu-

ture states. For example, a chatbot fed with examples of text chats can learn to converse with humans in real-world applications.

AI programming focuses on three cognitive skills: learning, reasoning and self-correction.

Learning Processes

AI programs focus on acquiring data and creating rules for turning that data into actionable information. Rules, also known as algorithms, provide step-by-step instructions to complete a specific task.

Reasoning Processes

The success of an AI program depends on choosing the right algorithm to reach the desired outcome.

Self-Correction Processes

AI programs are designed to continually enhance their algorithms to provide the most accurate results.

Case Study: Why is artificial intelligence important?

AI allows business organizations to gain insights into their operations that they may not have been previously aware of. There are numerous tasks that AI-enabled machines can perform better than humans. This has acted as the driving force for exploring new business opportunities. It was because of AI that computer software could be used to connect riders to taxis. Uber, one of the largest companies in the world, is utilizing sophisticated machine learning algorithms to predict when people would need a greater number of rides in a particular area so that the company can get drivers on the road before they are needed.

1.1.2 Advantages and Disadvantages of Artificial Intelligence

Artificial intelligence, when used along with machine learning and deep learning techniques, is quickly evolving to processes massive amounts of data much faster and makes predictions more accurately than humanly

possible. While huge volumes of data are being generated every day, only AI applications can utilize this data to quickly turn it into actionable information.

Google has become one of the largest players in AI. It provides a range of online services by using machine learning to understand how people use their services and then improving them.

Advantages

1. Performs well on tasks that uses detailed data.
2. Takes less time to perform tasks that needs to process huge volumes of data.
3. Generates consistent and accurate results.
4. Can be used 24 X 7.
5. Optimizes tasks by better utilizing resources.
6. Automates complex processes.
7. Minimizes downtime by predicting maintenance needs.
8. Enables companies to produce new products having better quality and speed.

Disadvantages

1. Involves more cost.
2. Technical expertise required to develop and use AI applications.
3. Lack of trained professionals.
4. Incomplete or inaccurate data may result in disastrous results.
5. Lacks the capability to generalize tasks.

In 2017, Google's CEO, Sundar Pichai, pronounced that Google would operate as an 'AI first' company.

Conclusion: Today, every big enterprise has used AI to improve its operations and gain advantage on their competitors.

1.2 History of Artificial Intelligence

Let us analyse the support for the modern field of AI, right from 1943 to the present day (**Fig. 1.1**).

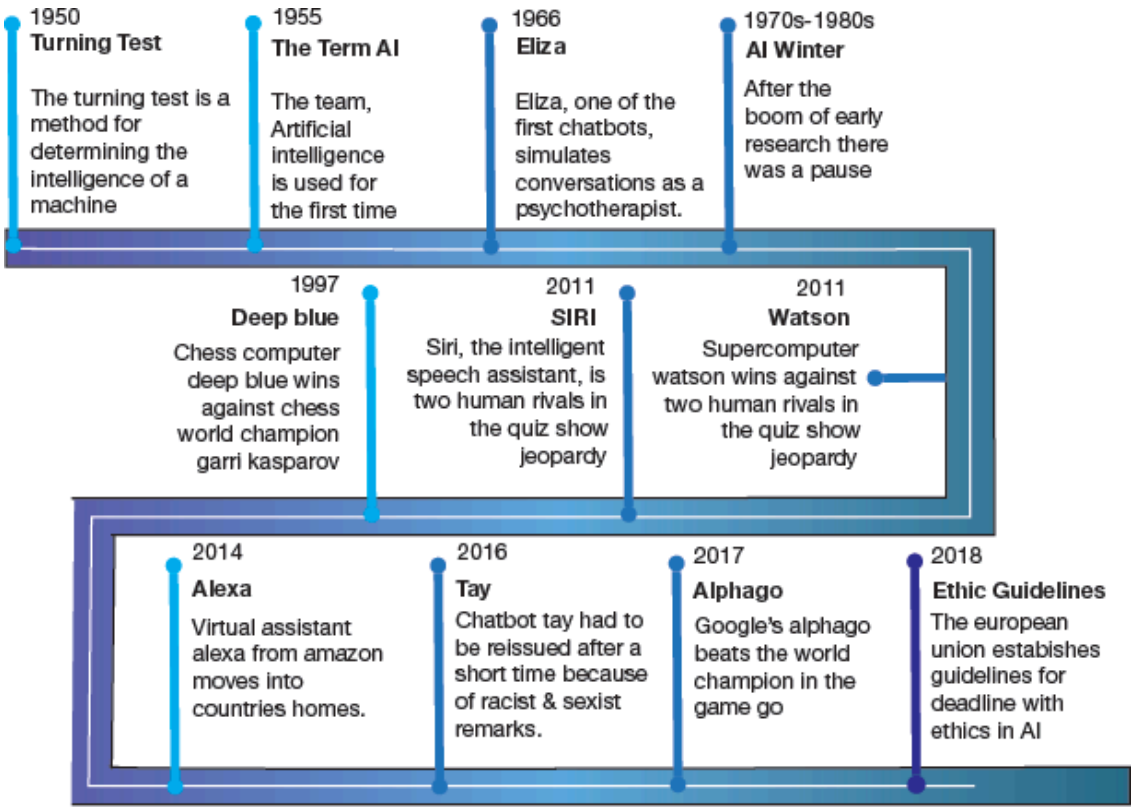


FIGURE 1.1 History of Aritificial Intelligence (AI)

In **1943**, Warren McCullough and Walter Pitts wrote a paper that proposed the first mathematical model for building a neural network.

1950 was a significant year in the field of AI. A series of events took place this year including the following:

- Alan Turing demonstrated Turing Test, that could determine whether a machine is intelligent or not.
- Harvard undergraduates Marvin Minsky and Dean Edmonds built the first neural network computer.
- Claude Shannon publishes a paper on programming a computer for playing chess.
- Isaac Asimov published his work on the ‘Three Laws of Robotics.’

In **1952**, Arthur Samuel developed a self-learning program to play checkers.

In **1954**, the IBM computer could translate 60 carefully selected Russian sentences into English.

In **1956**, John McCarthy coined the term ‘artificial intelligence’. The scope and goals of AI were discussed in a conference later. It is from this point that we consider the birth of artificial intelligence as we know it today.

In the same year, Allen Newell and Herbert Simon demonstrated the first reasoning program.

In **1958**, John McCarthy developed the AI programming language, Lisp. He also published a paper discussing a complete AI system that could learn from experience as effectively as humans do.

In **1959**, Allen Newell, Herbert Simon and J.C. Shaw developed the General Problem Solver (GPS), a program designed to imitate human problem-solving.

In the same year, Arthur Samuel coined the term ‘machine learning’.

In **1963**, John McCarthy started the AI Lab at Stanford.

MIT Professor Joseph Weizenbaum developed ELIZA, an early natural language processing program that laid the foundation for today’s chatbots.

In **1969**, the first successful expert systems to diagnose blood infections was developed at Stanford.

In **1972**, logic programming language PROLOG was created.

The 1974–1980 period is known as the ‘First AI Winter,’ because it was during this period that DARPA cutbacks in academic grants and funding for AI dried up, stalling further research.

In **1980**, DEC developed the first successful commercial expert system, R1.

In **1982** Japan launched the ambitious Fifth Generation Computer Systems (FGCS) project to develop supercomputers that could support performance and a platform for AI development. Later, in **1983**, even the US government funded research in advanced computing and artificial intelligence.

1987–1993 marked the 'Second AI Winter.' In 1992, Japan terminated the FGCS project owing to failure in meeting its ambitious goals. The US also ended the project in 1993 after spending nearly $1 billion as the results were far from what were expected.

In **1997**, IBM's Deep Blue machine could beat world chess champion, Gary Kasparov.

In **2005**, STANLEY, a self-driving car, won the DARPA Grand Challenge. Even the US military started investing in autonomous robots like Boston Dynamics' 'Big Dog' and iRobot's 'PackBot.'

In **2008**, Google did a remarkable work in speech recognition and introduced the feature in its iPhone app.

In **2011**, Apple released Siri, an AI-powered virtual assistant that could be used through its iOS operating system.

In **2012**, Andrew Ng, founder of the Google Brain Deep Learning project, fed 10 million YouTube videos as a training set to a neural network and used deep learning algorithms to enable the neural network to recognize a cat without being told what a cat is.

In **2014**, Google makes the first self-driving car that could pass a state driving test. In the same year, Amazon released Alexa.

In 2015, Baidu's Minwa supercomputer used a convolutional neural network (special deep neural network) to identify and categorize images with a higher rate of accuracy than the average human.

In **2016** Google DeepMind's AlphaGo defeated world champion Go player Lee Sedol. Go is an ancient Chinese game and winning this game was a

major breakthrough in the field of AI.

In the same year, the first 'robot citizen', a humanoid robot, Sophia was created that could perform facial recognition, verbal communication and facial expression.

In **2018**, Google released NLP engine BERT that reduced barriers in translation and understanding by machine learning applications. In the same year, Waymo One service was launched that allowed users to request a pick-up from one of the company's self-driving vehicles.

In **2020**, Baidu released its LinearFold AI algorithm to develop a vaccine during the early stages of the SARS-CoV-2 pandemic. The algorithm could predict the RNA sequence of the virus in just 27 seconds, 120 times faster than other methods.

1.3 Types of Artificial Intelligence

We can classify an artificial intelligent system into one of the following categories (**Fig. 1.2**).

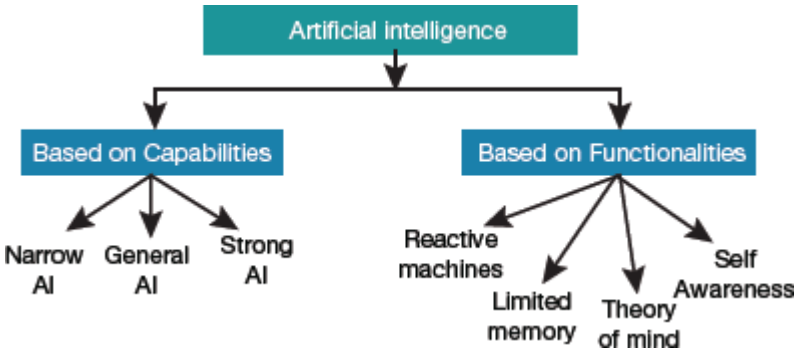


FIGURE 1.2 Types of Artificial Intelligence

1.3.1 Weak AI

Weak AI, also known as narrow AI,

is specifically designed to perform a specific type of task. For example, Siri and Alexa are weak AI systems. These systems are already trained with appropriate responses to classify things accordingly. When you instruct Alexa to play a song, it responds by playing that song.

In fact, majority of AI applications that we use today (predicting weather, stock prices, optimizing business, etc.) come under this category. Weak AI systems operate within a limited context and are the most successful realization of artificial intelligence to date. Application of narrow AI has resulted in significant societal benefits. Google search, Image recognition software, self-driving cars and IBM's Watson are some examples of such systems.

1.3.2 Strong AI

Strong AI, also known as artificial general intelligence or artificial super intelligence (ASI) or superintelligence, makes full attempt to resemble the human brain. It utilizes cognitive skills and fuzzy logic to perform tasks for which it had not been trained earlier. Such a system needs capability for visual perception, speech recognition, decision-making and translations between languages.

As of now, we see general AI only in sci-fi movies, where machines emulate human intelligence and think strategically to perform complex tasks. However, it is believed that soon, ASI will surpass the intelligence and ability of the human brain as demonstrated by HAL, the superhuman, rogue computer assistant in *2001: A Space Odyssey.*

Many people fear that intelligent robots would overrun humanity, but experts says that we need not worry about it as it is not likely to happen in the near future.

Moreover, artificial intelligence systems can also be categorized **into** four groups based on their functionality**.** This categorization can be given as discussed in **Table 1.1**.

TABLE 1.1 Weak AI vs Strong AI

Weak AI	Strong AI
It supports a narrow range of applications with a limited scope.	It supports a wider range of application with a wide scope.
This application is good at specific tasks.	This application has an incredible human-level intelligence.
It uses supervised and unsupervised learning to process data.	It uses clustering and association techniques to process data.
Example: Siri, Alexa.	Example: Advanced robotics

1.3.3 Reactive Machines

Reactive machines are very basic machines with no memory to store and thus use past experiences to determine future actions. They just perceive the world and react to it. IBM's Deep Blue, which defeated chess grandmaster Kasporov, is such an example. The AI system sees the pieces on a chess board and responds without referring to any of its prior experiences. Since it lacks capabilities to store past experiences, a reactive machine cannot improve with practice.

A strong AI program must pass a Turing Test as well as the Chinese room test.

Reactive machines can perform only a limited number of specialized tasks. Such machines are more trustworthy and reliable as they react the same way to the same stimuli every time.

Deep Blue could only identify the pieces on a chess board and compute its next move based on the rules of chess and the present positions. The AI application could not pursue future potential moves by its opponent or plan its own pieces in a better position. Every move was treated as its own reality, without any connection with any other previous movement.

Google's AlphaGo is another example of a game-playing reactive machine. The application evaluates future moves using neural network. This gives it an edge over Deep Blue game.

1.3.4 Limited Memory

Limited memory machines retain data but only for a short period of time and can use it only for a limited time. They cannot keep adding data permanently to a library of their experiences. Such systems are usually used in autonomous vehicles since they need not store data (like *recent speed of nearby cars, distance between cars, the speed limit, etc.) that can help them navigate roads*.

AlphaGo also bested world-class competitors of the game, defeating champion Go player Lee Sedol in 2016.

No doubt, limited memory AI is more complex and is used for better possibilities than reactive machines. These machines continuously train a model to analyse and utilize new data. The model also improves using feedback received from humans or the environment and stored as data.

There are three major machine learning models that applies limited memory artificial intelligence:

1. **Reinforcement learning:** It enables continuous learning through repeated trial-and-error to make better predictions.
2. **Long Short-term Memory (LSTM):** Such models use past data to help predict the next item in a sequence. More recent information is given a higher priority for prediction than past data. However, this does not mean that past data is not considered while making predictions.
3. **Evolutionary Generative Adversarial Networks (E-GAN):** The ML model evolves over time to explore new ways of utilizing previous experiences to jump formulate new decisions. The model is continuously striving to discover a better path. It makes use of simulations and statistics, to predict outcomes throughout its evolutionary mutation cycle.

1.3.5 Theory of Mind

The theory of mind focuses on imitating the human brain. For this, it forms representations about the world, starting from thoughts, emotions

and memories. However, such systems are only theoretical as of now but they may become reality very soon.

These machines must understand thoughts and emotions that affect the behaviour of one's self. Theory of Mind machines make decisions keeping in view the feelings experienced through self-reflection and determination.

1.3.6 Self-Awareness

Once Theory of Mind is established, then the next step would be incorporating self-awareness in AI systems. Self-awareness in machines enable them to possess human-level consciousness, understand the current state —its own existence in the world and use that information to deduce others feelings. Self-AI systems will be able to understand what others may need. They could interpret user's feelings by learning not only what they communicate to them but also how they communicate it. Knowledge imbibed by researchers and its learning of the conscious context will help them to respond to an event. However, such systems do not yet exist but may become available in the near future.

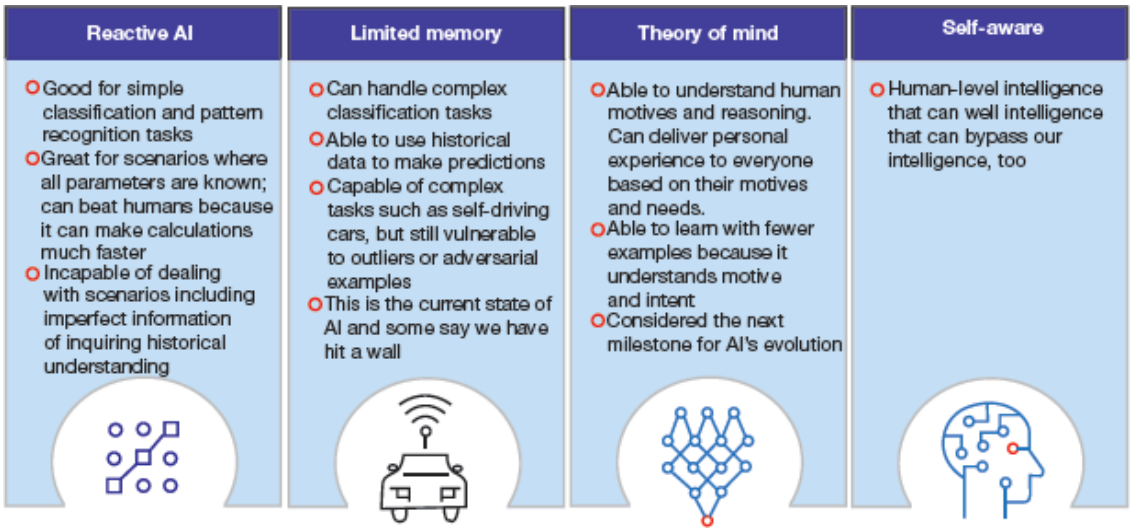


FIGURE 1.3 Types of AI Systems Based on Functionalities

1.4 Is Artificial Intelligence Same as Augmented Intelligence and Cognitive Computing?

Some experts think that *artificial intelligence* is the same as augmented intelligence. But this is not true. Let us understand the difference between

the two terms.

Augmented intelligence is weak AI that simply improves products and services. For example, automatically highlighting vital information in business is a part of implementing augmented intelligence.

Artificial intelligence or true AI, strong AI or artificial general intelligence, is the future AI that would far surpasses the human brain's ability to understand and implement complex tasks. As of now, though lot of work is being done in this domain, AI remains within the realm of science fiction. Technologies like quantum computing can play an important role in making AGI a reality.

The term AI when used in reference to machines, makes machines obtain human intelligence by simulating how humans sense, learn, process and react to information in the environment.

Cognitive computing is a term that is especially coined for products and services that mimic that augments human thought processes.

MACHINE LEARNING

Let us try to understand the term 'machine learning' with the help of some real-world scenarios.

Scenario 1: Imagine that you are given the task of finding the missing number in the series-10,20,30,40, ?

Yes, you are correct. That's 50. But how did you get the answer?

You quickly did some computations in your mind and observed the pattern in the data values. The whole idea of machine learning is to imbibe the same thinking process in machines.

Scenario 2: Before going for playing a cricket match, you always use your past experiences to understand how a batsman/bowler plays. For example, if you know a bowler gives a straight ball after every 2 left balls and 1 right ball, then you can better use this information to prepare yourself for

the next ball. In machine learning, we make machines also to learn from past experience/data.

Scenario 3: When you buy a new mobile phone, you can yourself compute the price of that phone vis-à-vis the configuration and brand. For example, you know a phone with 128 GB RAM, stereo speakers, iOS 16, hexacore would cost around 80K. So, given the details, we can predict the price. Similarly, a machine learning program is developed to predict values based on certain specifications.

Now let us go deeper in this terminology. Machine Learning is a branch of computer science that analyses data and identifies patterns to teach a machine to deduce results and make decisions without any human intervention.

ML algorithms learn from experiences rather than instructions.

They automatically learn and improve by learning from their output. For this, humans do not have to write instructions for them to produce the desired output. They learn by analysing data sets and comparing the final output. In case of any error, they repeat the learning process until the accuracy of the outputs improve.

This automation not only saves human time and effort but also makes better decisions.

Thus, machine learning is an application of artificial intelligence (refer **Fig. 1.4**) that enables machines to automatically learn and improve from experiences without being explicitly programmed.

It enables computers (machines) to make data-driven decisions rather than being explicitly programmed for carrying out a certain task. These algorithms are designed to learn and improve over time when are exposed to new data. Thus, the process of teaching the machines is a structural process that can be broken down into three steps (**Fig. 1.5**).

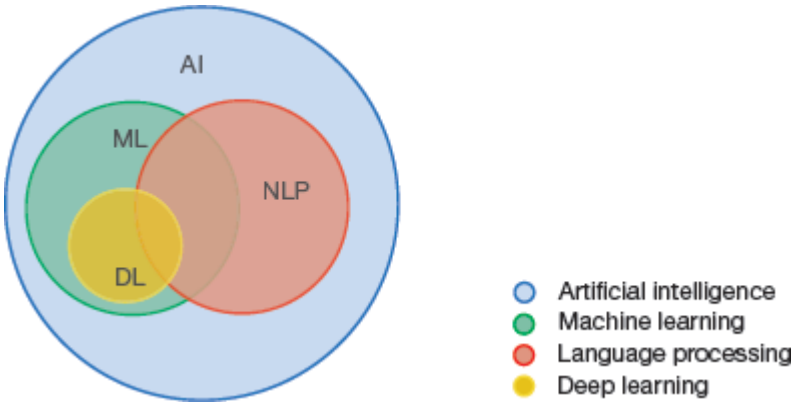


FIGURE 1.4 Relationship between Artificial Intelligence, Machine Learning, Deep Learning and Natural Language Processing

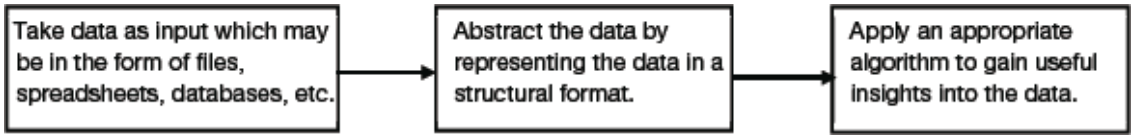


FIGURE 1.5 Making the machines learn

HOW IS AI RELATED TO MACHINE LEARNING?

Machine learning and artificial intelligence are two different techniques where machine learning is a tool for achieving artificial intelligence. The aim of AI is to create intelligent machines that can recognize human speech, have vision to see, assimilate knowledge, strategize and solve problems as humans do.

Machine learning provides machines the ability to learn, forecast and progress on their own without specifically being programmed. ML system improves by learning from more data and experiences.

Imagine you have been given the task of making a robot see, talk, walk, sense and learn. Which technique will you use—ML or AI? Of course, AI. This is because to achieve this task, multiple technologies come into picture (like NLP, computer vision, voice recognition, reinforcement learning, etc). Machine learning is just one of them that will be used only for learning purpose. Therefore, AI is the superset and ML is the subset of AI. That is, ML is only a way to achieve AI (**Fig. 1.6**).

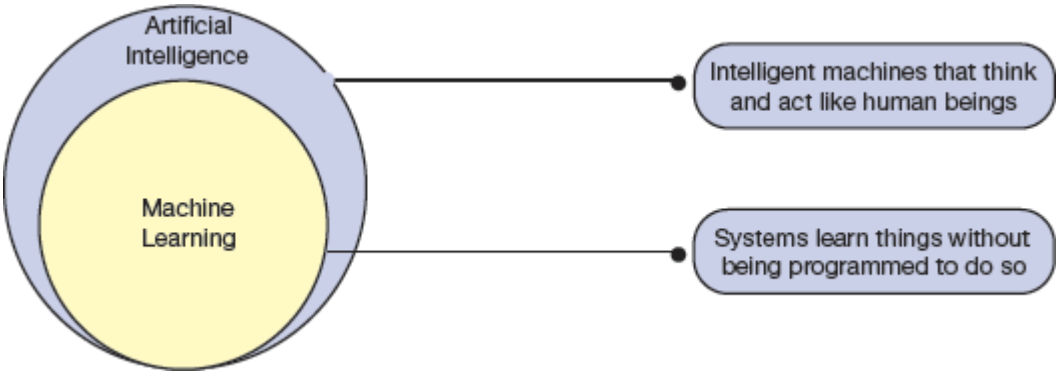


FIGURE 1.6 Difference between AI and ML

Traditional Programming vs Machine Learning

Traditional programming involves manually creating a program that accepts data and returns an output. It rests on a manual process as one has to manually formulate or code rules.



Traditional programming uses conventional procedural language like assembly language or a high-level language such as C, C++, Java, JavaScript, Python, etc. These languages code the rules and write lines of code manually. The program accepts input data, and based on the programming logic, generates the desired output.

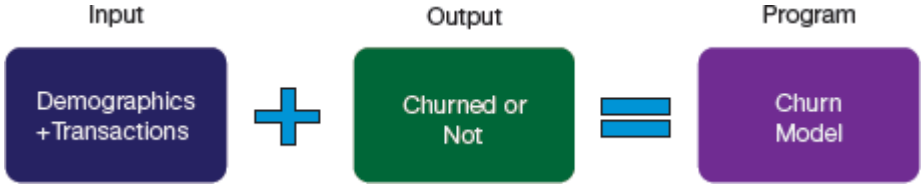
The conventional programming approach is algorithm dependent, and for a program, multiple algorithms can be used. The programmer can analyse the performance of all the algorithms to choose the best that meets the current requirements.

Recently, an advanced type of programming has revolutionized business by adding power to the applications working on intelligence and embedded analytics. **Machine learning** programming, also known as augmented analytics, accepts both data as well as output. It then generates the corresponding program for giving the specified output on that input. This gives insights that can be used to predict future outcomes.

The machine learning algorithm automatically formulates the rules from the data. Unlike traditional programming, machine learning is an automated process that adds the capability of embedded analytics in areas including natural language interfaces, automatic outlier detection, recommendations, etc. All of these features give better insights and reduce decision bias.



For example, if we give customer demographics and transactions as input and use historical customer churn rates as output, then the algorithm will formulate a program (also known as **predictive model)** that can predict if a customer will churn or not.



Therefore, the steps to create a model to predict outcomes can be given as follows:

Step 1: Identify the business question you would like to ask.

Step 2: Identify the historical input.

Step 3: Identify the historically observed output (when the condition is true and when it is false).

Machine learning language goes beyond traditional algorithmic solutions and trains the machine to solve different complex tasks by itself.

Unlike conventional programming, machine learning uses a pre-written algorithm that learns how to solve the problem itself. It is a more sophisticated way of solving a problem.

For Programmers: A data engineer cannot work as a substitute to a conventional programmer and vice versa.

Every data engineer must know at least one coding language. However, a conventional programmer may not learn the machine learning algorithms.

The capabilities of both traditional programming and machine learning are altogether different and cannot replace each other (refer **Table 1.2**). However, machine learning supplements conventional programming. For example, while machine learning can build predictive algorithms, traditional programming, on the other hand, can be used to design user interface or perform data visualization.

Case Study: It is not easy to filter Amna's photos from a set of 100 images using traditional programming. Even comparing every pixel value may not give accurate results. But the same task can be easily done with machine learning. We just need to input different photos of Amna, the machine learning model will learn how she looks like.

TABLE 1.2

Traditional Software	Machine Learning
The primary objective is to meet functional and non-functional requirements.	The primary goal is to optimize the performance of the model as every 0.1 % improvement result in significant business value creation.
The quality of the software primary depends on the quality of the code.	The quality of the model depends upon various parameters related to the input data and hyperparameters tuning.
Traditional software is created using traditional programming languages like C, C++, Java.	Machine learning models can be created using different algorithms. Each of these algorithms may perform differently.
As compared to machine learning, few parameters are required.	To get an accurate prediction, a large number of parameters may be required.
Change in data does not change/impact the business functionality. Only new business rules may have to be accommodated.	Machine learning models may have to be retrained with every new data to ensure same or better performance.
Example: Look at the steps given below to convert Celsius scale to Fahrenheit scale Step -1: Take input (Celsius) Step-2: Apply the conversion formula: Fahrenheit = Celsius * 1.8 + 32 Step -3: Print the Output (Fahrenheit) Here, the programmers tell the computer what to do on the input data (multiply Celsius with 1.8 and add 32) to get the result.	Example: Look at the steps given below. Step 1: Feed values in Celsius (i.e., -40, -10, 0, 8, 15, 22, 38) Step -2: Feed corresponding Fahrenheit values (-40, 14, 32, 46, 59, 72, 100) Step -3: Pass these 2 sets of values to Machine Learning (ML) model Step- 4: The ML model computes (convert) the logic to predict any other Fahrenheit value from the given Celsius value. The ML model automatically generates the relationship between Celsius and Fahrenheit to do the conversion.

1.5 Machine Learning and Deep Learning

Majority of AI systems are supported by breakthroughs in machine learning and deep learning techniques. These techniques together go hand-in-hand to deliver an efficient system that sometimes it is just difficult to understand the difference between artificial intelligence, machine learning and deep learning. However, venture capitalist Frank Chen highlighted this difference by stating *'Artificial intelligence is a set of algorithms and intelligence to try to mimic human intelligence. Machine learning is one of them, and deep learning is one of those machine learning techniques.'*

In simple words, machine learning techniques are used to feed data in to the computer so that statistical techniques can be applied to it to help machines 'learn' how to solve tasks or which it had not been specifically pro-

grammed. These computers free programmers from writing several lines of code. Machine learning techniques can be further classified into supervised learning (using labelled data sets) or unsupervised learning (using unlabelled data sets).

Deep learning is an advanced machine learning technique that passes data inputs through a biologically-inspired multiple layers of neural network. Such ***a neural network architecture consists of a number of hidden layers through which the data is processed.*** Data processed in this architecture allows machines to go 'deep' in its learning, making connections and weighting input for the best results.

How does AI work?

AI algorithms solve specific tasks in the following manner:

1. Studying very large amounts of data.
2. Making generalizations or statistical estimations model (Refer **Fig. 1.7**).

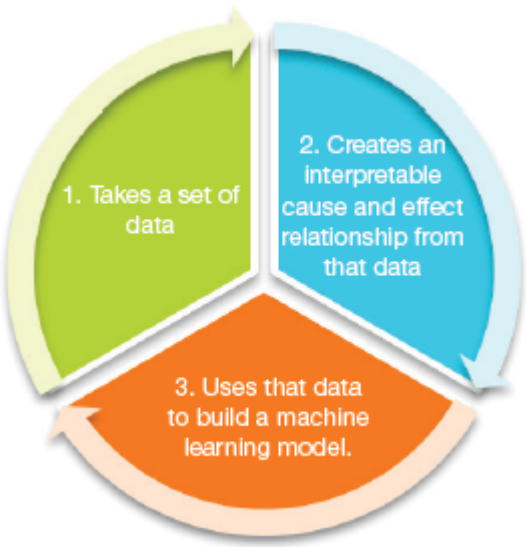


FIGURE 1.7 Building a ML model

Features of AI

1. Artificial intelligence is **autonomous** and can make independent decisions without any human intervention. These systems work silently in the background without the user's knowledge. The best part of an AI

system is that they don't have to be programmed manually to respond to every situation. Rather, these systems themselves learn through input data and past experiences.

Baidu, a search engine that works in the Chinese language, has a voice cloning tool, which can clone human voice with just 3-4 seconds of audio, as opposed to the previously taken 30 minutes

2. P**redict and adapt: AI systems can** understand data patterns to make decisions and predictions.
3. An AI system **continuously learns** from patterns in data.
4. An AI system is said to be **reactive as i**t perceives a problem and acts on perception.
5. **Data driven:** In the last several years, the cost of technology has gone down considerably. With the availability of cheap data storage (hard disk, etc.), fast processors (CPU, GPU or TPU) and sophisticated deep learning algorithms, huge volumes of data can now be easily extracted. This has led to the rise of data centric AI systems.
6. Data driven AI systems can make accurate predictions based on past experiences have outperformed humans. However, success of these systems depends largely on the availability of correctly labelled large datasets.
7. Finally, an AI system is **futuristic** since the scope of it being applied in different areas is continuously expanding.

1.6 Applications of AI

Artificial intelligence has become the cornerstone of innovation. When AI is used along with machine learning and deep learning techniques, it can easily analyse and interpret abundance of data available to make useful predictions and/or automate excessively complex or mundane tasks.

AI technology is helping organizations perceive the world as humans do and then react to real-time events. Most of the companies across different areas are making the best use of AI while avoiding its implications to create an AI ecosystem in which the following are observed:

Business analysts and data scientists work together to define the problems and its objectives.

Data engineers manage the data and the underlying technology to make it operational for analysis.

Data scientists plan, develop, analyse, visualize and model data using a machine learning model.

IT architects take care of the underlying infrastructure required for supporting data science projects in the cloud or on the organization's servers.

Application developers deploy the models into applications to build data-driven products.

In big and small organizations, AI is being used in the following areas.

1. **Automation:** When used in automation tools, AI technologies can enhance the number, efficiency and types of tasks performed. For example, Robotic Process Automation (RPA), is a software that automates repetitive, rules-based data processing tasks traditionally done by humans.
2. **Machine vision:** As the name suggests, this technology allows machines to see. For this, the machine captures visual information using a camera and then analyses using analog-to-digital conversion and digital signal processing techniques. Unlike human eyesight, machine vision is not bounded by biology. For example, it can be programmed to see through walls. Nowadays, machine vision is used in various applications ranging from signature identification to medical image analysis.
   Machine vision or computer vision facilitates computers to derive meaningful information from digital images, videos and other visual inputs. Information gained through these inputs is then used to take action. When used with convolutional neural networks, computer vision is used in applications involving photo tagging in social media, radiology imaging in healthcare and self-driving cars in the automotive industry.
3. **Recommendation systems:** A recommendation system recommends or suggests products, services, information to users after analysing data based on a number of factors including history, behaviour, preference and interests of the user. Such systems are an example of data-

driven AI as they need a lot of data for training. For example, you have seen that when you watch a video on YouTube, it suggests many videos that match the videos that you generally search for, prefer to watch or have been watching in the past. In this case, the data about your preferences and history of your activities and behaviour are analysed. Similarly, online shopping websites (like Ajio, Flipkart, Amazon, etc.) recommends you products that you normally prefer or buy (refer **Fig. 1.8**).

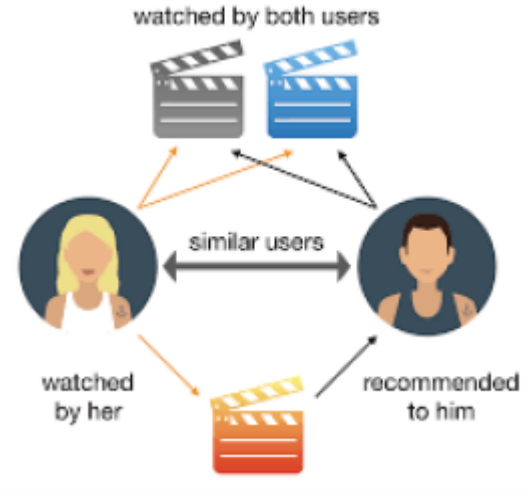


FIGURE 1.8 Recommendation System

4. **Natural language processing (NLP):** Enabling machines to understand and respond to commands given in human language (in digital assistants) has become possible because of NLP—the capability in machine to process natural language (refer **Fig. 1.9**). Another popular example of NLP is spam detection, which looks at the subject line and text of an email and decides if it is a junk mail or not. Practically, NLP

techniques come under machine learning.

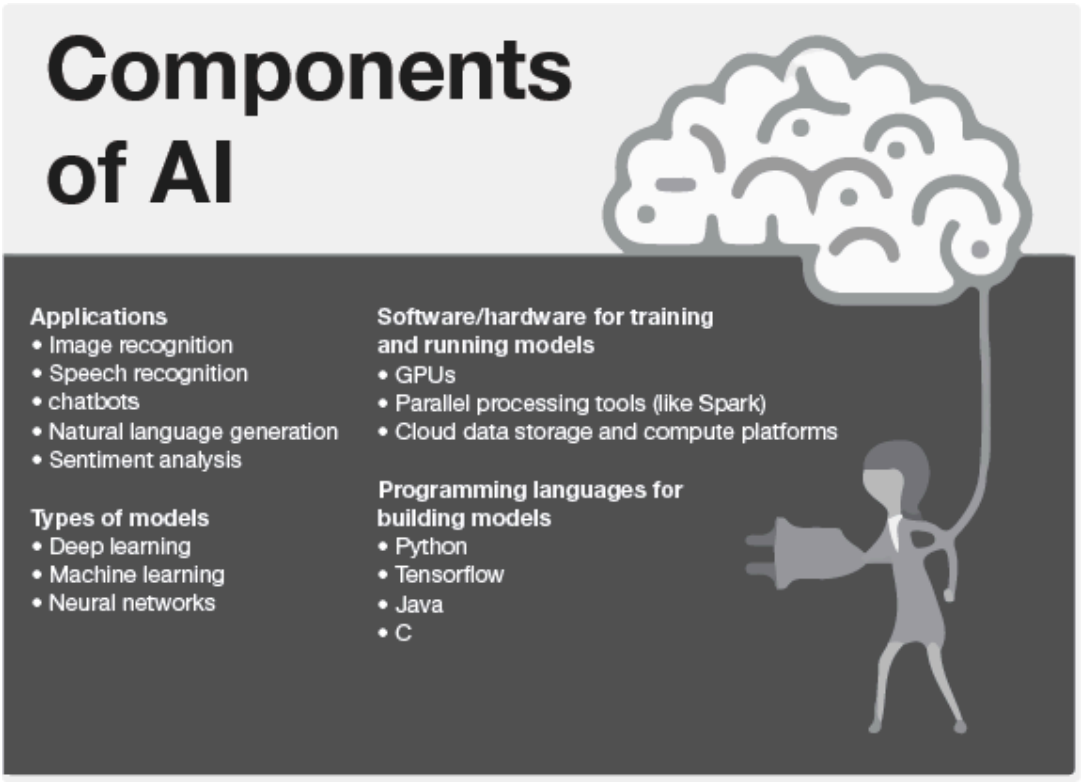


FIGURE 1.9 Components of AI

All smartphones today use voice-to-text feature with which users can start speaking and the phone converts the audio into text. You must have observed that you can ask Google to search a particular keyword.

5. **Robotics:** Robotics is the science of designing and manufacturing robots. Robots are often used to perform tasks that are either risky for humans or require great precision. Industrial robots now function as cobots performing smaller, multiple tasks while collaborating with humans in warehouses, factory floors and other workspaces.
   For example, robots are used in assembly lines for car production, by space scientists to move large objects in space or to click images of areas deep under the sea. These days, extensive research is being done to make robots that can interact in social settings.
6. **Self-driving cars:** Autonomous vehicles or driverless cars use a combination of techniques including computer vision, image recognition and deep learning to drive a vehicle while staying in a given lane and avoiding unexpected obstructions, such as pedestrians.
   The AI system in these vehicles collects data from the vehicle's radar, cameras, GPS and cloud services to generate control signals required to operate the vehicle. From Google's self-driving car Waymo LLC to

Tesla's 'autopilot' feature, AI has revolutionized the automotive industry.

How self-driving cars work? Autonomous vehicles have LIDARS (light detection and ranging) and remote sensors installed that gather information from the vehicle's surroundings. LIDAR uses light from a radar to see objects in front of and around the vehicle. It interprets this information to quickly determine the presence of objects, distances between vehicle and the object. On-board computers combine this information with the data from the sensors to compute the likelihood of the vehicle hitting something. In case the vehicle experiences dangerous conditions, it would either shift lanes, slow down or stop completely.

7. **Personalized online shopping:** AI is extensively being used these days to provide users a personalized experience by showing them products based on multiple filters and using recommender system (that displays them products similar to what they had been searching for or they had purchased in the past).
   You must have observed that when you buy a product, say smartphone from an online shopping website, then AI algorithms on that site will automatically suggest various add-on products for the same model such as screen guards, back covers, earphones, compatible pairable devices, extended warranty plans, damage protection plans and much more.
8. **Enhanced images:** These days, cameras and apps use AI to apply a range of effects on images to enhance their quality, add lighting, blur out the background, apply filters, identify scope of the picture and capture details to let users tag their friends and family.
9. **Social media:** These days, social media is generating an immeasurable amount of data through chats, tweets, posts, etc. In this context, AI is used for applications involving face verification and detecting facial features. To do so, every small detail from an image is extracted and analysed. Moreover, every time you access Facebook, you must have observed that news feed pushes articles based on your interests.
10. **Surveillance:** Traditionally, images from security cameras are monitored manually. Even a small error in this task may be detrimental. In such a scenario, an application based on AI can be developed that takes input from security cameras and trains the system using super-

vised learning algorithms. Such a system can identify potential threats (like intrusion, invalid access, unidentified individuals on defined premises, etc.) and generate an alarm for human security officers to investigate further. This maturing technology is playing a big role in helping organizations fight off cyber-attacks.

The DRDO has developed a robot named ***Daksh*** that safely destroys life-threatening objects. This is a big step to safeguard our soldiers in combat scenarios. Robots are also deployed in anti-submarine operations, fire support, battle damage management, strike missions and laying machines.

11. **Agriculture:** With the increased demand for food owing to population growth, climate change and concerns for food security concerns, organizations are shifting towards automation and use of robotics technology to help farmers explore better ways to protect their crops against bad weather, weeds, market consumption rates, etc. With image recognition capability, AI systems can be used to capture images of crops through farmer's camera and then identify possible defects in them. Early detection of potential threats and then providing soil restoration techniques, tips, and other possible solutions to deal with the identified defects have proved to be very beneficial to farmers.
12. **Prevent natural disasters:** AI systems can be fed with data about thousands of previous disasters. This can help it to accurately predict the occurrence of that disaster in future. For example, some AI systems are being specifically designed to study data about previously occurred earthquakes and similar disasters like tremors and volcanic eruptions, to create a ***neural network*****.**

    Similarly, AI systems are also fed with **seismic information (**information regarding movements in the tectonic plates that reside beneath the Earth's surface) and data about ash particles that arise from lava after a volcanic eruption, and other geological information to predict sudden eruptions.

    Other AI systems are designed to accurately predict other natural disasters like cyclones, floods, etc. Did you know that cyclone Fani hit Bangladesh and the south eastern state of India, Odisha in May 2019? The Indian Meteorological Department predicted and kept a stringent vigilance on the situation and evacuated about 1.2 million residents of Odisha on time, and moved them to cyclone shelters built on higher al-

titudes. This saved countless lives. Surprisingly, the death toll due to the cyclone was only 72 people.

13. **Customer service:** These days, businesses are cautious to retain existing customers and also acquire new customer base. You must have observed that almost every website/application has a provision for customers to chat with its representatives. Chat support bots or chatbots are AI systems that use NLP to understand and then extract information from the site and present it to the customer as and when required. These chatbots answer frequently asked questions (FAQs) around topics, like shipping, or provide personalized advice, cross-selling products, etc. Messaging bots on e-commerce sites with virtual agents, messaging apps, such as Slack and Facebook Messenger are all examples of chatbots.

    Virtual assistants like Siri, Cortana, Alexa, Amazon's Echo translate human language into desirable actions using NLP. NLP allows them to user's commands (to control other devices, book cabs, make phone calls, order food, etc.) given in user's language.
14. **Video games:** With AI, the gaming industry has reached new heights. The AI system can play with human counterparts and defeat them easily. They are trained to take optimized decisions at every step and also learn from their mistakes to avoid repeating the same mistake again in future. Besides learning, AI is also used for designing the game, developing the characters and for framing the story to a certain extent (refer **Fig. 1.10**).



Credit: franz12 / Shutterstock

FIGURE 1.10 Playing games in a virtual world

15. **Healthcare:** The capability of AI to crunch numbers fast and learn from historical data, makes this technology very useful in the healthcare industry. Detailed analysis has fostered research and quick decision-making. Pathologists are using AI to analyse samples of tissue to diagnose the problem more accurately.
    Moreover, chatbots that can be used as digital health assistants ensure proper medication and treatment of patients. Such systems can help patients find medical information, schedule appointments, understand the billing process and complete other administrative processes. Thus, AI systems have been doing well to improve patient outcomes and reduce costs. For example, IBM Watson is one of the best-known healthcare technologies that understands natural language to respond to questions asked of it. The system also mines patient data and other available data sources to form a hypothesis, and then evaluate it with a confidence scoring schema.
    An array of AI technologies including drones are being used to fight against pandemics such as COVID-19. Apart from providing personalized medicine and X-ray readings, personal health care assistants are used to remind patients to take their medicines, exercise or eat healthier.
    Robots are also used in laparoscopy, neurosurgery, orthopedic surgery, disinfecting rooms, giving medicines to patients and for lot more other tasks.
16. **Banks:** Banks and other financial institutions are using AI-based systems to provide customer support, detect anomalies and credit card fraud. AI systems can also analyse data to provide financial advice and suggest stocks to buy. Some AI systems are specifically designed to optimize stock portfolios. They can make thousands or even millions of trades per day without human intervention.
    Banks are also using chatbots to give information about their services and offerings and to customers so that they can easily handle transactions that do not require human intervention. They are also using AI to improve their decision-making for loans, and to set credit limits and identify investment opportunities.
17. **Smart homes:** More and more tech savvy people today are increasingly using numerous smart devices like smart locks, switches, refrigera-

tors, washing machines, etc. to experience the comfort of a smart home.

Smart devices that make homes smart, learn user's behaviour patterns to save their time and resources, and money by implementing cost-optimized operations to provide a smarter way of living. For example, thermostats and building management systems can be used to automate building heating and cooling. These systems learn and can then predict when to turn on/off the boiler based on weather conditions inside as well outside ethe house.

18. **Travel:** Travel companies are using AI to provide customer support, analyse historical and real-time data to determine prices to different destinations, evaluate the level of traffic on the roads to give an accurate estimated time to reach a particular point that too through different modes of travel, including bus, trains and flights.
19. **Space exploration:** Space expeditions and discoveries require analysing vast amounts of data using machine learning and deep learning techniques. For example, astronomers used AI to interpret data obtained by the Kepler telescope to identify a distant eight-planet solar system.

    AI was used by NASA's rover mission to Mars to perform investigations on Mars.
20. **Mining:** In mining, robotics is used in applications like robotic dozing, excavation and haulage, robotic mapping and surveying, robotic drilling and explosive handling, etc. A mining robot can easily navigate flooded passages and use cameras and other sensors to detect valuable minerals. Robots can also be used to protect humans from any harm or injury in the excavation process by detecting gases and other materials that are a potential threat to humans.
21. **Education:** Robo-readers are used in schools to evaluate answer sheets and grade them. Students and teachers are also notified about the topics in which a particular student is doing well as well as topics in which he/ she needs more practice.

    Banking organizations use AI to improve their decision-making for loans, and to set credit limits and identify investment opportunities. From microbiological research to archaeological advancements, all these tasks that were difficult earlier have become much easier today

due to AI. And the scope will only grow with time. Applications that utilize Virtual Reality (VR), training simulations and exploration technologies are already paving the way for building a better future.

22. **Law:** In the legal field, AI is used to automate labour-intensive processes to save time and improve client service. Machine learning techniques can be used to describe data and predict outcomes. In addition, computer vision can be used to classify and extract information from documents and NLP can be used to interpret requests to provide the requested information.
23. **Retail and manufacturing:** Even in showrooms, AI is being used to provide virtual shopping capabilities that offer personalized recommendations, make the customers aware about different payment options available to them, manage stock, etc. AI units deployed in manufacturing units can be fed with data streaming from IoT devices to forecast expected load and demand.
    In the manufacturing sector, robots are used for cutting, welding, assembly, disassembly, organizing printed circuit boards, packaging, labelling, palletizing, product inspection, testing, colour coating, drilling, polishing and handling materials.
    AI can be used everywhere where there is a need to increase productivity and profitability by reducing human efforts. The industrial robots are accurate, flexible, provide less expensive labour cost, generate less noise, give less defects, minimize damages and considerably improve productivity.

1.7 Robotics—an Application of AI

Robotics is a branch of engineering that involves the conception, design, manufacture and operation of robots. It is an inter-disciplinary field that includes the study of electronics, computer science, artificial intelligence, mechatronics, nanotechnology and bioengineering.

Science-fiction author Isaac Asimov, first used the term ‘robotics’ in the 1940s. According to him, robots must ensure the following:

Rule 1: Never harm humans.

Rule 2: Always follow instructions from humans without violating rule 1.

Rule 3: Protect themselves without violating the other rules.

Some important aspects of robots include the following:

1. They have electrical components for providing power and control the machinery.
2. They have mechanical construction and their shape or design depends on the task they are intended to accomplish.
3. They are programmed. Instructions fed to them helps them to determine what, when and how it would do something.

These days, robots are being extensively designed as bots that explore the Earth's harshest conditions and assist in almost every facet of healthcare. Though the robotics industry is still evolving, we already have robots that work in the deepest depths of our oceans to the highest heights of outer space. They are being used to do everything that humans could not dream of doing.

1.7.1 Types of Robots

Pre-programmed robots operate in a controlled environment to perform simple, monotonous tasks. For example, a mechanical arm on an automotive assembly line that is used to weld a door, or insert a certain part into the engine. This type of robot can perform that task for longer hours, faster and more efficiently than a human.

Humanoid robots look like humans and/or mimic human behaviour. They can perform human-like activities (like running, jumping and carrying objects). Sophia and Atlas are two popular examples of humanoid robots.

Autonomous robots operate without human operators. They are designed to perform tasks in open environments that do not require human supervision. For example, the Roomba vacuum cleaner uses sensors to roam freely throughout a home to clean it.

Teleoperated robots are mechanical bots controlled by humans. They are used in extreme geographical conditions, weather, circumstances, etc.

For example, human-controlled submarines used to fix underwater pipe leaks during the BP oil spill or drones used to detect landmines on a battlefield are examples of teleoperated robots.

Robots are also used for tasks varying from diffusing bombs to performing surgeries. VR robots are also becoming popular these days.

Augmenting robots either enhance current human capabilities or replace the capabilities a human may have lost. Robotic prosthetic limbs or exoskeletons used to lift hefty weights are examples of such robots.

1.7.2 Uses of Robotics

Robots are used in the following areas.

1. Helping fight forest fires
2. Assisting humans in manufacturing plants (known as co-bots)
3. Provide companionship to the elderly
4. Work as surgical assistants
5. Deliver food orders or other packages
6. Perform household tasks like vacuuming and mowing the grass
7. Locate and transfer items throughout warehouses
8. Perform search-and-rescue missions after natural disasters
9. Landmine detectors in war zones

Industry-wise applications can be given as follows:

The manufacturing industry is amongst the first well-known users of robots. They use robots and co-bots (bots that work alongside humans) to efficiently test and assemble products, like cars and industrial machinery.

Logistics companies use robots in their warehouses to perform tasks like shipping, handling goods and ensuring quality control. Robots are supposed to take items off the shelves, transport them across the warehouse floor and package them. They are now also used for last-mile delivery of packages for faster and efficient delivery.

Self-driving cars are a result of integrating data science with robotics. Automakers like Tesla, Ford, Waymo, Volkswagen and BMW are all working to provide users an ultimate experience of travel that will let them sit back, relax and enjoy the ride. Companies like Uber and Lyft are also developing autonomous rideshare vehicles that will be operated without humans.

The healthcare industry uses robots to perform complicated surgeries, deliver everything from medicines to clean linens.

Space agencies like NASA use robots in different ways. Robotic arms on spacecraft can move large objects in space. Robotic spacecraft can visit other worlds like the Moon or Mars. For example, Mars rovers Spirit and Opportunity are robots. Cassini studies Saturn and its Moons and rings. Robots like The Voyager and Pioneer spacecraft are now travelling beyond our solar system. People on the Earth use computers to send messages to the spacecraft. The robots have antennas that pick up the message commands and work as per the instructions given to it.

Robotic airplanes can fly without a pilot aboard. NASA is also developing Robonaut to help people in space. The upper body of Robonaut looks like a person. It has a chest, head and arms. They can work outside a spacecraft and work like an astronaut on a spacewalk.

NASA is also working on robots that might help an astronaut in an emergency. For example, when the astronaut is seriously hurt, a doctor on Earth could use the robotic arm to perform surgery. This technology can help doctors on Earth, as well. Doctors can help people in faraway places where there are no surgeons to perform complicated surgeries.

Robots are also used as **scouts** to check out new areas to be explored. They take photographs, measure the terrain, look for dangers and find the best places to walk, drive or stop. This not only helps scientists and engineers make better plans for exploring but also helps astronauts to work more safely and quickly.

Note:

- ASIMO is a humanoid that has the ability to recognise moving objects, postures, gestures, understand its environment, and interact with humans.
- Pepper is the world's first robot capable of recognising human emotions. It is social, can converse with people, give them directions and even dancing with them
- Romeo helps with everyday tasks, assist when people have fallen over, make conversations and play games.
- Buddy is designed to entertain the family, help you with your everyday activities, offer reminders, provides recipes in the kitchen, make video calls, keep an eye on your home while not at home, connect all your smart home devices together and even help your children learn.
- Panasonic Egg uses NLP Robot to communicate with you. It's an intelligent assistant that can be controlled by your voice, play video footage via a built- in projector and even engage in interactive games. This robot is Wi-Fi connected and promises software updates in future to improve it further.
- REEM is a full-size humanoid service robot that can act as a receptionist, provide entertainment for guests, make presentations and give speeches in different languages and help with a variety of different chores. REEM is able to self-navigate, interact with people it encounters and keep on running for up to eight hours.

1.8 Drones Using AI

The term 'drone' generally means any unpiloted aircraft that operates using a combination of technologies including computer vision, artificial intelligence, object avoidance tech and others. Also known as 'unmanned aerial vehicles' (UAVs), these drones can be as large as an aircraft or as small as the palm of your hand. With drones becoming readily accessible, they are now increasingly being used for the most dangerous and high-paying jobs that are discussed below.



Credit: Slavoljub Pantelic / Shutterstock

FIGURE 1.11

Emergency Response

Drones outfitted with thermal imaging cameras are used by emergency response teams to identify victims who are difficult to spot with the naked eye.

Humanitarian Aid and Disaster Relief

During times of natural disaster, drones are used to assess damage, locate victims and deliver aid. In addition, they are also used to prevent disasters altogether (refer **Fig. 1.14**).

Surveillance

Drones outfitted with thermal imaging cameras can be used to monitor and combat forest fires. Thermal cameras can measure and detect abnormal forest temperatures. This information can then be used to identify areas that are more prone to forest fires or identify fires just minutes after they begin (refer **Fig. 1.13**).

Conservation

Poaching and activities resulting in climate change have adversely affected wildlife worldwide. In fact, according to the World Wildlife Fund, thousands of species are estimated to extinct each year. To help combat

this trend, conservationists are extensively using drones for geospatial imagery to monitor and track animals to protect our biological ecosystem.

Disease Control

Many infectious diseases spread through animals. In such a scenario, drones can be used to capture and test mosquitoes for infectious disease. This initiative can not only protect local residents, but can also prevent epidemics before they begin.

Moreover, drones are also being used in remote areas to provide quick access to vital drugs, medicines and medical equipment. All these initiatives have a profound impact on preventing disease, increasing life expectancy and raising general standards of living.

Bomb Detection

Small sized drones fitted with effective cameras can easily penetrate into constricted spaces to detect live bombs and save lives of thousands of people.

Air Strikes

Drones are used for conducting air strikes. Former US President Barack Obama had been using drones regularly to attack militants in the tribal areas of Pakistan. While being controlled by the defense personnel, drones can be made to fly around suspected areas to fulfill military operations (as shown in **Fig. 1.12**). However, use of drones for military operations has also raised numerous moral and ethical concerns as they lack accountability and failure to fully grasp the consequences of actions.



Credit: Alex Yuzhakov / Shutterstock

FIGURE 1.12 Usig Drones for Air Strikes

Agriculture

Farmers in some advanced countries are extensively using drones to gather data, automate redundant processes and improve efficiency to reduce costs and expand yields. Drones also help farmers to predict their potential harvest.

Weather Forecasting

Scientists are using drones to collect data about temperature, humidity, wind speed and other climatic parameters that could help them to accurately predict future changes to global weather systems. Drones used as autonomous sailboats are used to collect oceanic and atmospheric data from the ocean surface.

Maritime

Drones are used to inspect ships above the surface as well as hulls from below. Countries like the Netherlands, Denmark, and Norway are already using drones to identify ships committing emissions infractions.

Waste Management

Drones are being used to clean oceans, collect waste in ports and harbours and maintain systems for wastewater management.

Energy

Drones are used by energy generating companies to set up new sites for the production of energy. For this, the drones are made to survey areas and gather topographic detail that can be used to help oil and gas companies identify new drill sites. Drones are also used to extract, refine and transport oil and gas while ensuring compliance with regulations and standards.

Drones fitted with specialized thermal sensors can detect leaks faster than a human inspector. Even companies generating solar energy can use drones to design configurations for new arrays.

Mining

Mining activities require constant measurement and assessment of stockpiles of ore or rock or minerals. Drones fitted with unique cameras can capture large amounts of data from the air, thereby reducing the risks associated with having surveyors on the ground.

Drones are being using by the Indian government to deliver COVID vaccines even in the remotest areas.

Construction Planning

Drones are used to improve construction planning, enhance project monitoring and site management. Cameras fitted in these drones monitor buildings and gauge topography and soil type throughout the construction lifecycle.

Urban Planning

With increasing urbanization, cities are overburdened to accommodate more people in already congested spaces. In such a scenario, urban planning is being done with drones to implement data-driven improvements. For example, drones are used to gather data in populated areas. This data when analysed using ML algorithms can suggest areas that may get bene-

fited from green space or classify the different types of structures and regions on the map.



Credit: 102 / Shutterstock

FIGURE 1.13 Drones for Surveliance

Personal Transportation

China-based EHANG was started as an autonomous aerial vehicle (AAV) that operates with four rotors (quadcopter) for vertical takeoff. The vehicle is used to help passengers reach their destinations. Such a vehicle is especially very useful in an urban environment with plenty of obstacles. Personal transportation drones require minimal inputs from the passenger and aims to allow safe landings even in case of engine failure or a collision.

Big companies like Uber, Airbus, Boeing and Rolls-Royce are working on developing flying drones (robotaxis) for ferrying passengers around.

Space

Drones are being used to be moved in the space. For example, NASA is using a drone-like helicopter in its Mars 2020 mission to help look for signs of life on Mars. In this mission, the helicopter will act as a scout for the rover, gathering data about the planet's terrain and surveying areas the rover cannot reach.

NASA is also using a nuclear-powered drone for exploring Titan, one of Saturn's moons. The drone will arrive on Titan by 2034, autonomously traverse the planet for about 2 years, taking photos and sending data back for analysis.



Credit: koya979 / Shutterstock

FIGURE 1.14 Drones for delivering medical help

Telecommunications

Telecommunication towers must be inspected frequently to ensure service reliability. However, this process is too dangerous and time-consuming to do manually. In such a scenario, drones are able to quickly assess damage to help guide repair teams in restoring service.

Internet

Drones are being used to provide Internet access in remote areas and over irregular landforms. For example, Facebook has designed a solar-powered drone called Aquila, to provide Internet access to rural parts of the world. However, in 2018, Facebook halted the use of Aquila and used only third-party drones instead.

Even SoftBank, in collaboration with the drone manufacturer AeroVironment, is planning to develop drones that will operate in the stratosphere to serve as ‘floating cell towers’ to provide internet service to customers.

Outdoors

Drones are used outdoors to perform aerial landscape photography and extreme sports footage and map the entire mountain face to help climbers and skiers to better understand the terrain.

Tourism and Hospitality

These days, flying drones providing luxury accommodations can be used to travel to new locations on demand or to remote and traditionally inaccessible locations for guests. They are also being used to deliver packages and room service quickly.

Live Entertainment

Drones are already being used by Disney for entertainment through synchronized lights shows, floating projection screens and as drone puppeteers.

Journalism and News Coverage

News companies are using drones to gather news, especially from areas that are difficult to visit due to safety issues, high costs or physical barriers. For example, drones are used to get aerial footage of the aftermath of hurricane, wildfires and assess flood-ridden areas in the midwest.

Food Services

Drones are being used by online food ordering and delivery services for faster, cheaper delivery. This helps restaurants downsize their physical locations and lower real estate expenses.

1.9 The Future of AI

Accenture presented a report, AI: Built to Scale, in which it was concluded that 84% of business executives believed that they required AI to achieve a higher growth but 76% struggled to use AI across their business. When asked about the future of AI, Dr Kai-Fu Lee, AI oracle and venture capitalist said, in 2018, ‘[AI] is going to change the world more than anything in the history of mankind. More than electricity.’

Today, the most important topic that restrains companies to go deep into AI is cybersecurity. Spike in cybersecurity breaches throughout 2020 when attacks rose 600% is a big concern. During the pandemic, hackers capitalized on people working from home, using less secure technology

and Wi-Fi networks. In such a scenario, AI and machine learning can be used to identify and predict potential threats and catch instances of fraud especially those involving finances. Moreover, every application, these days, collect a lot of data about users. A person with disruptive mindset can train a machine to even threaten user(s) with solutions that can even endanger their being.

AI is also considered to be a threat to privacy as it violates human rights. For example, voice assistants and facial recognition techniques invade privacy of humans and probably eavesdropping into their lives. Thus, AI needs to evolve as a trustworthy system in the future.

AI will change the way managers search suitable candidates for a job. AI is already being used in the hiring process so much that up to 75% of resumes are rejected by an automated Applicant Tracking System (ATS) before even reaching a human being.

In the past, recruiters have to invest considerable time to analyse resumes and shortlist relevant candidates. But now, this is easily done by AI-powered programs. However, the concern is to develop AI systems that are not biased. HireVue is a startup (established in 2018) that used facial recognition software and psychology to determine the potential effectiveness of a candidate in a certain role. But after the Electronic Privacy Information Center filed a lawsuit alleging that it could perpetuate bias and prejudice, HireVue had to discontinue use of facial recognition software in early 2021, and now uses audio analysis and natural language processing. This clearly indicates that the use of AI in the hiring process or any other process may become controversial.

As of now, AI systems like **Jobscan** is an excellent resource that provides similar resume scanning by comparing a candidate's resume to a job description, thereby performing better than the ATS. Similarly, **Jobseer,** a browser add-on, is an AI-based tool which scans a resume, as well as keywords and skills related to the desired jobs, to match with the job listings that best fit a candidate's experience. For each listing, the candidate gets a rating as well as recommendations of skills to be added to improve the capabilities.

AI and machine learning are one of the most desired skills in today's job market. Jobs demanding AI or machine-learning skills are expected to increase by 71% in the next five years. Even an understanding of how AI can be used in different sectors is considered to be an added skill. Some facts and myths that prevail about this technology are listed in **Fig. 1.15**.

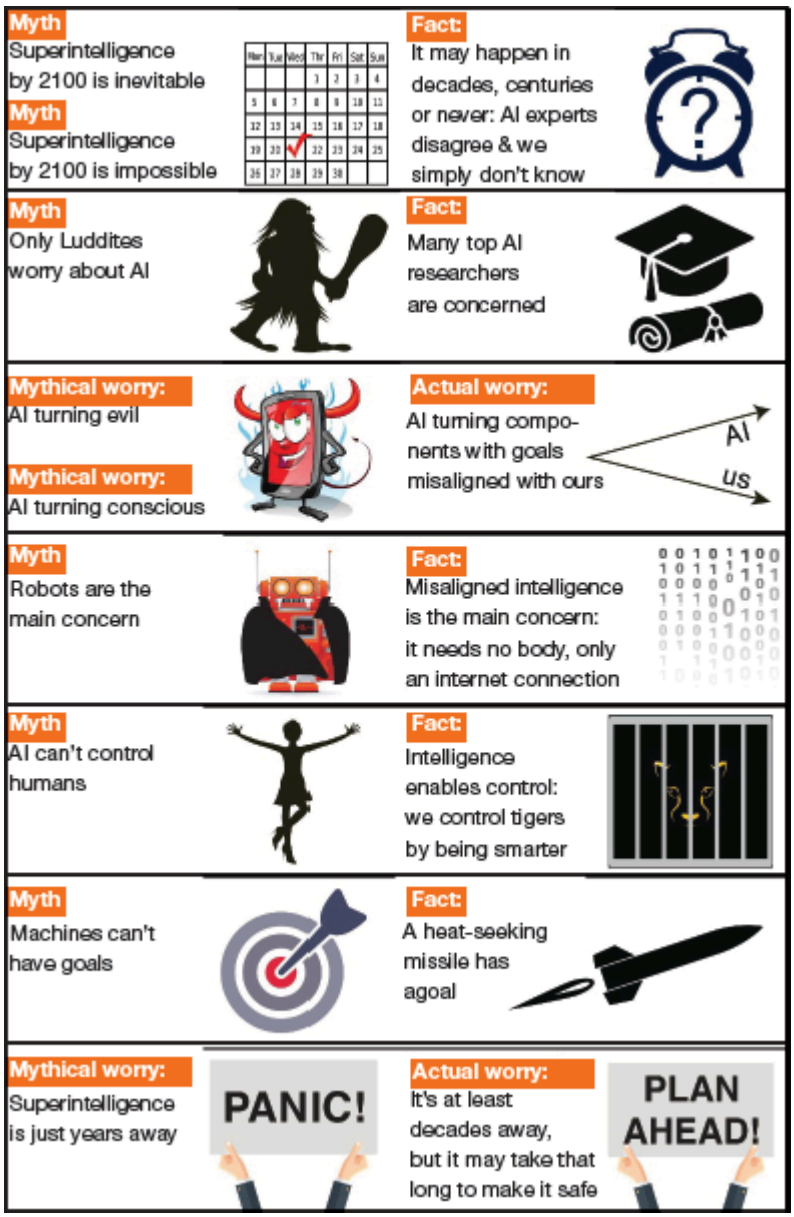


FIGURE 1.15 Interesting Facts and Myths about AI

AI technology is still evolving. Companies would strive to implement ethical AI or responsible AI. Business organizations are becoming keener to earn better revenues and profits. Small businesses are considering AI solutions of larger companies as a case study to understand how they can delve into it at a lower cost for maximizing their success. Startups are coming up to offer agility, growth and better ideas to stay competitive.

In a few years' time, advances in AI will reach the superintelligence stage. But it should be adopted without harming the goodwill of loyal people. No doubt, the future of artificial intelligence is exciting and extremely promising.

In fact, recently, a paper published by researchers at AMOLF's Soft Robotic Matter group, discussed self-learning robots that can easily adapt to changing circumstances. For this, small robotic units were connected to each other for them to learn to move on their own. Techniques like reinforcement learning and generative adversarial networks will be in great demand. Moreover, sustainable technologies for fighting climate change will be developed using AI. The term already coined for this is 'Green AI'.

To conclude, AI as a technology is not at all a threat to human existence. But the real threat is from humans who may misuse AI to harm an individual or a society. For example, robotic army can be fed to realize harmful motives. In order to alleviate the potential risks of AI in the future, AI systems must be ethical and responsible.

1.10 No Code AI

Right from the first generation of computers, exhaustive efforts have been made to make programming easier, faster, less technical so that it can become accessible to a much larger audience.

No-code tools allow people to build applications and systems without having to write long and complex program codes. Applications can now be designed using visual interfaces and guided user actions. No-code tools are pre-integrated with other tools to exchange information as needed. Given below are some applications that can be built entirely using such tools.

1. Websites and landing pages with Webflow.
2. Web or mobile applications with Bubble, Adalo, Mendix or Thunkable

   Number and applications of No Code AI tools will grow steadily
3. Chatbots or virtual assistants through Octane AI, Kore.ai, Landbot or mindsay

4. Databases through Airtable
5. Connecting your tool stack with Zapier, tray.io, Integromat, Parabola or Paragon
6. E-commerce through Shopify or Weebly
7. Manage memberships with Memberstack
8. Newsletters with Mailchimp or Mailjet

1.10.1 Why No-Code AI?

It has been seen that majority of businesses struggle to implement AI to its full potential and scale as it is still considered to be a core-engineer's task. In such a scenario, use of no code AI tools can be a great help due to the following reasons:

1. These tools allow automation through plug-and-play or drag-and-drop UI.
2. Even users with absolutely no coding skills can leverage the benefits of AI algorithms to solve business issues.
3. Businesses that cannot afford to invest additional time and resources to develop AI applications can build such systems from the ground up.
4. **Improves accessibility** to intensify use of data science or AI by even small and mid-sized companies.
5. **Enhances usability** as more and more non-technical users can create AI based solutions to a problem in low-cost and less-time.
6. No-code AI solutions are of fairly good quality and reduces human error(s) when setting up such systems. Refer **Fig. 1.16** to know about some No Code AI Tools. **Figure 1.17** list their applicabilities.

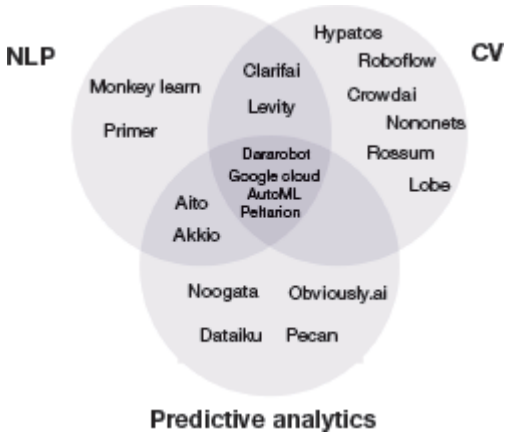


FIGURE 1.16 No-code AI landscape 2022: Technology

	Text	Image	Docs	Video	Audio	Tabular
No Code						
Levity	✓	✓	✓			
Aito	✓					✓
Akkio	✓					✓
Lobe		✓				
Monkey learn	✓					
Obviously.ai						✓
Pecan						✓
Low code						
Clarifai	✓	✓		✓	✓	
Crowdai		✓			✓	
Dataiku						✓
Dararobot	✓	✓		✓	✓	✓
Google cloud AutoML	✓	✓		✓		✓
Hypatos			✓			
Nononets			✓			
Noogata						✓
Peltarion	✓	✓			✓	✓
Primer	✓					
Roboflow		✓		✓		✓
Rossum			✓			

FIGURE 1.17 No-Code AI landscape 2022 : Data type

When running this exercise, it made sense to group the tools by the technology they use. Here's a quick round of definitions:

Computer Vision

It allows machines to obtain information from digital images, videos, pdfs and other visual data, and take actions based on their learnings.

NLP

This allows machines to understand and process language both spoken and written, for example, text messages.

Predictive Analytics

This refers to predictive modelling based on structured (i.e., tabular data), for example, predicting churn rate, forecasting and stock prices.

We also made a distinction between no-code tools and low-code tools. The no-code tools follow basic criteria: they are end-to-end tools that are usable without coding knowledge. In that sense, low-code tools are better suited if you have someone on your team who *speaks data.* **Figure 1.18** categorizes these tools based on their usage.

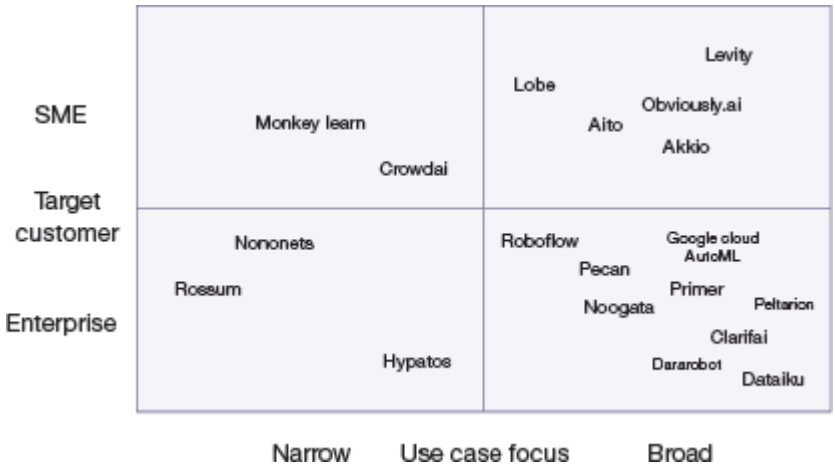


FIGURE 1.18 No-code AI landscape 2022: Customer vs use case

Let us look at some popularly used no-code tools (**Table 1.3**).

TABLE 1.3 No-Code AI Tools and their Usage

Tool	Description
Aito	Performs predictive analytics and NLP automation Has a simple UI Furnished with APIs that integrate well with many automation platforms Focus on tabular datasets Offers automated re-training system Usually designed for sales, marketing and financial activities
Clarifai	An NLP and Computer Vision tool Modelling unstructured data for the entire AI lifecycle Offers image, video and text recognition solutions
Crowd AI	Uses Computer Vision for both images and video Used for applications like vegetation management or disaster responses
Levity	Focuses on image, text and document classification Enables users to train custom models on their use-case-specific data automatically learn from user interactions Easily integrates with all the tools
Google AutoML	Trains high-quality models specific to business needs AutoML Image for image annotation and labeLling AutoML Video for video annotation and labelLing. AutoML Text for semantic analysis and text classification
Lobe	Free, private desktop application by Microsoft Used for image classification, with object detection, and data classification coming soon Has pre-trained solutions (e.g., offers emotional reactions to enable the program to react to different expressions) Allows people to send emoji reactions using just their faces
MonkeyLearn	Performs text analysis Facilitates data visualization Supports unstructured text-based data Used to get topic, sentiment, intent, keywords, etc. Also used for automatically tagging business data, visualizing actionable insights and trends Provides simplified processes for text classification as well as extraction
Nanonets	Used in the domain of Computer Vision Used to build an ID card verification model for any country, format or language
Noogata	Predictive analytics tool Helps to customize your model Used for making data-driven decisions
ObviouslyAI	Uses NLP processing to perform tasks on user-specific text data. DragS and drop CSV file or integrate with HubSpot, Salesforce, or MySQL for getting predictions Performs tasks like forecasting company revenue, optimizing supply chain, personalizing marketing campaigns, etc.
Primer	NLP model builder Powerful integrations with other tools Offers pre-trained models that are ready to be used Offers features to visualize your model performance in one go
Apteo	Helps to make more repeat sales AnalySes customer database to find out which products certain customers are

	likely to buy next Using this information, crafts better messages for more accurate targeted upselling It integrates with ecommerce tools like Shopify, Stripe and Square
Prevision	Increases the productivity of data science projects Focused on AI modelLing Users can upload data, train the model, analyse performance and then get predictions
causaLens	Finds cause-and-effect relationships in data Uses an innovative approach to AutoML
Causaly	Same as causal AI field, but is focused on finding causal relationships in biomedical science research Uses machine learning to analyze over 30 million papers, clinical trials and side effect databases
PredictNow.ai	A vertical specific no-code AI solution focused on financial machine learning Computes the probability of profit for your next investment
Accern	Functions for credit risk, ESG investing, quantitative research, financial crimes analysis, etc. For all applications involving heavy-duty finance needs
RunwayML	Made for creators and creatives, with features for image, video and text data Allow users to create synthetic images and videos, cut objects out of videos and more
IBM Watson	Helps in model creation, training, and deployment, without any serious integration features
MLJar	Highly focused on modelling, automating feature engineering, algorithm selection, documentation and explanations
Dataiku	Supports visual AutoML functionality More focused on general data science tasks like data integration, building data pipelines, data visualizations and statistical analysis Provides users with a drag-and-drop interface for more advanced users, Dataiku also provides the ability to code in Python and R within the platform Offers a wide range of tools for data preparation, machine learning and deployment For data preparation, Dataiku offers an interface for ETL tasks like data cleaning, transformation and enrichment
Noogata	Has pre-built AI blocks Used for predictive maintenance, fraud detection, and customer segmentation. Users can simply drag and drop these blocks to create an AI solution
Kinetix	Uses no-code AI to create high-quality 3D animations Users provide any video, which gets converted into 3D animated avatars In the Web3 world, these avatars can then be used for a variety of purposes, such as in video games, social media or even as customer service agents
Loris	Provides conversational AI software to help human agents make customer support more human Help businesses automate customer support tasks with a high-touch, human experience Crafts real-time suggested responses based on the customer's tone and sentiment
Neptune Software	Uses AI-augmented workflow to support users in creating apps, either from scratch or from a library of customizable templates and components
Evisort	Embeds AI into contract management software

	Helps businesses automate the contract management process, from data extraction to analysis and approvals Uses AI to understand the contents of contracts, as well as to identify risks and opportunities
Webio	Allows businesses to quickly use conversational AI Integrates with SMS, WhatsApp and other channels, to allow communication with business customers

1.10.2 Future of No Code AI

Businesses are steadily moving towards no-code platforms. Research shows that by 2024, nearly 65% of application development will be done through either through low-code or no-code platforms. And, no code AI will play a big role in this. Some advantages of these tools are as follows.

1. No-code AI is a code-free technology.

   Google AutoML is the no-code AI solution that provides a feature-rich suite of AI products that helps to train high-quality ML models tailored to unique business needs.
2. It facilitates non-AI experts also to implement and test their ideas without any help of AI experts
3. AI experts can create ML solutions in less time using minimum efforts.
4. It supports collaboration between AI experts and domain experts.
5. It has drag-and-drop or wizard-based interface to build applications.
6. It helps to solve problems with improved productivity and efficiency.

All in all, no code AI ensures that **AI technology can be adopted by everyone and everywhere.**

1.10.3 Why No-Code AI Must be Used?

No code AI (refer **Fig. 1.19**) will become more important as it solves the following problems.

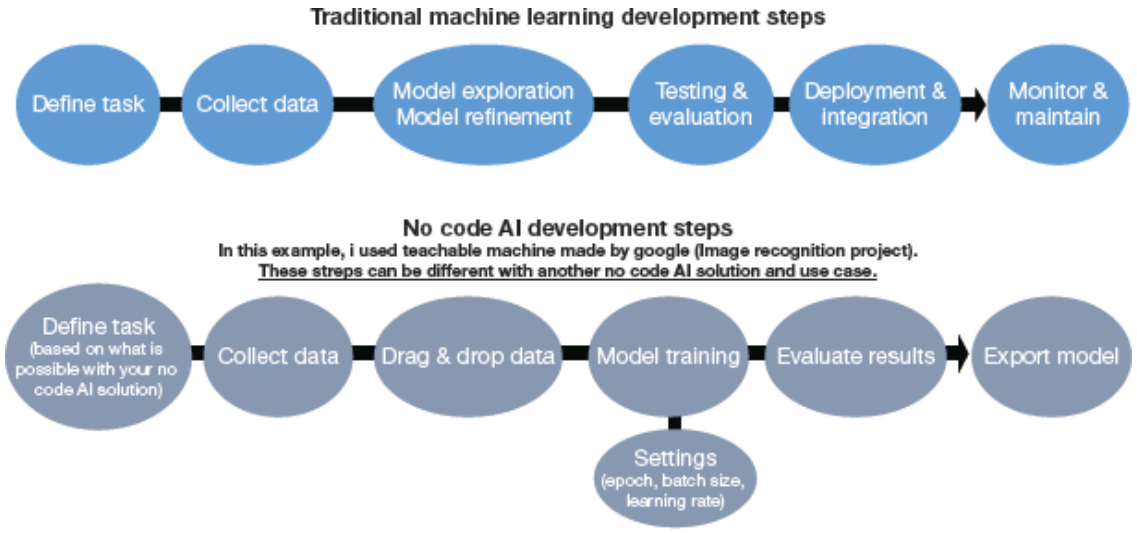


FIGURE 1.19 Steps in a Traditional ML Model Development and in a No-Code AI Model Development

Gap Between AI Experts and Domain Experts

Every year, more and more companies are adopting AI technology. There is a huge gap between AI experts and domain experts. They rarely understand terminology used by others. While AI experts talks using terms of machine learning (ML) models and optimization algorithms, domain experts, on the other hand, use only business-specific words to **solve their problems**. This gap is critical to the success of AI initiatives. Many projects fail due to lack of understanding. In such a scenario, no-code AI platforms allow domain experts to test their ideas and communicate their issues with AI experts in a better way. At the same time, these tools also help AI experts to add value for the domain experts in less time with less effort.

We can categorize no-code AI platforms based on the type of interface: drag-and-drop or wizard-based.

Limited AI Experts and More Problems

We know that many problems can be solved fully or partially using AI technology. But to solve them, we need AI experts. These experts are still

less in number and charge a heavy fee. So, here, domain experts can run their experiments and get AI solutions using No-code AI platforms.

Steve Jobs said, 'the line of code that's the fastest to write, that never breaks, that doesn't need maintenance, is the line you never had to write.'

It is interesting to note that only 0.25% population of the world can do coding but anyone can be taught to quickly and easily use no-code tools.

1.11 Low Code AI

The use of no-code as well as low-code platforms is constantly on the rise. Today, the growing demand for AI solutions in business applications has paved the way for low-code and no-code AI tools. These tools give companies the flexibility and agility to create new applications.00:00

Low-code is a software development technique that was introduced in 2011 to promote application development in less time with little to no coding required. For this, it supports visual development of applications using intuitive modeling with a graphical user interface (GUI).

Low-code development platforms are based on the concepts of model-driven design, automatic code generation and visual programming. These platforms provide integrated tools to eliminate the need to write code line-by-line. When working on these platforms, the user draws flowcharts in a visual editor, and the code is produced automatically. Users can even create software applications by dragging and dropping pre-defined components. Today, some of the most popular low-code software platforms that are used by both programmers and non-programmers are Mendix, Outsystems, Creatio, Appian and Creator.

According to Gartner, the demand for information systems will increase five times faster than IT departments' ability to provide them because the number of employees is not growing at a sufficient pace.

Lower the difficulty level of application development processes, broader is the audience. Low-code solutions not only enables fast delivery of applications with minimum effort but also ensure least possible effort for the installation and configuration of environments, training and implementation. Apart from other reasons, low-code AI is also preferred as it provides built-in security and maintenance, saving costs and time. It also lowers development risks at a high return on investment.

The key differences between traditional and low-code development are given in **Table 1.4**.

TABLE 1.4 Differences between traditional and low-code development

Feature	Traditional Tool	Low-Code
Time to market	Slow to launch, and is slow to change	Quick to launch
Uses	Manual coding	Drag and drop design, pre-built components
Scope	Planned and developed as large, complex projects	Small and independent solutions
Development	Difficult to align to business demands and has slow turnaround times	Allows up to 5x quicker building, efficient testing, and higher ROI.
Maintenance	Requires additional development and involves expensive support	Development enables easy and quick updating or extending of the solution. This makes it excellent for prototyping
Integration	Software integrations are time and investment-intensive, documentation is required, as well as extensive testing phases	Offers pre-built connectors live debugging and creating web services with no coding
Deployment	Slow and complex; it requires multiple steps and development resources	offers one-touch deployment, an application is sent to production in just a single click
Low-risk/ high ROI	High risk is involved	robust security measures, data integration and cross-platform support are already built-in and can be easily customized

1.11.1 Low-Code vs No-Code AI Development

Low code development requires very less or minimal programming effort from the user, whereas no-code software does not require any program-

ming at all.

Though both low-code and no-code use building blocks to create applications, low-code facilitates the integration of custom software in the form of such building blocks.

1.11.2 Who Uses Low-Code Development?

Low-code development is beneficial for different types of users to speed up their projects and simplify the deployment process. Low-code AI platforms enable rapid development, testing and scaling of apps. In this section, we will discuss who use these platforms and for what purpose.

Low-Code for AI Beginners

Low-code and no-code AI platforms make AI technologies more accessible to technical and non-technical users. Businesses no longer need to hire a full-time AI developer. These platforms have made it much easier for beginners to get started with AI technologies.

Low-Code for Developers

Low-code platforms help AI developers (like Computer Vision Engineer) to develop AI applications, add their own custom software or AI models as integrations to the platform. This helps programmers and researchers to spend less time on repetitive coding tasks, write less code and focus resources on enhancing and optimizing the functionality of the application.

Low-Code for Researchers

Researchers can use low-code platforms to quickly develop prototypes, speed up their experiments and test each change without the need to invest in expensive infrastructure. Moreover, they can easily compare different approaches and benchmark multiple versions.

1.11.3 Low-Code AI Platform for Computer Vision

A low-code AI platform for automating computer vision applications is a difficult task. Lot of research is done in this area to overcome the chal-

lenges and technological problems. The latest AI low-code platform, specifically designed for computer vision applications is, viso.ai.

viso.ai is used to create custom computer vision and deep learning applications that process video feeds of numerous cameras in real-time with deployed AI algorithms. Using fine-tuned, pre-trained AI algorithms and pre-built modules in a visual editor, viso.ai quickly creates a custom AI vision solution. The platform continuously optimizes AI applications to achieve progressively better results.

1.11.4 Components of Low-Code AI Platforms

A low-code AI platform includes the following elements:

Graphical User Interface (GUI)

An intuitive GUI with clear visual elements can be easily used non-technical users to construct AI vision applications. The purpose and usage of the element must be easily understandable by the users.

Pre-Built Integrations

The AI vision application may use visual input of cameras and some pre-trained AI models, so these functionalities are available as ready to use building-blocks to the user. Pre-built integrations of modules that ensures secure data storage and communication are available with these platforms.

Application Manager

Developing an application with a low-code AI platform is just one portion of the software development process. The application needs testing, deployment and maintenance on a regular basis. For this, the platform offers tools to update code, remove bugs and improve its functionality and performance.

1.11.5 Disadvantages of Low-Code/No-Code Platforms

Though no-code/low-code AI platforms sounds interesting to use, there are some concerns which should be addressed before taking the final call in selecting a platform.

Security

Some platforms may not be able to design access protocols. In such a case, security may be compromised and this is a serious concern for many companies where security is at the utmost priority. Therefore, it is crucial to read, research and clearly understand the terms and conditions to interpret where and how data will be used.

Lack of Customization

Low-code/no-code platforms offer limited functionality as they are specifically designed to solve a particular problem.

Requires Consultation or Training

All users, technical or non-technical, must be able to use the low-code/no-code platforms. But usually, it is used by an ML engineer who also need lot of training and consultations to guide other team members to solve problems using AI.

Lack of Trust

According to Google Trends, the interest in no code ML is increasing but people interested in traditional ML are far ahead. Support for libraries for ML and computer vision are much more than low-code/no-code AI platforms.

Lock-In Strategy

Considerable amount of switching cost is involved while moving from one vendor to another. Thus, most of the times, a user is entirely dependent on a particular software vendor.

Limitations on Personalization

Some no-code and low-code solutions do not allow users to change certain parameters.

Data Management

Even while using no-code solutions, businesses may have to rely on the expertise of data scientists and data engineers for data processing tasks.

Scalability

As of now, creating scalable solutions for solving complex problems using a no-code machine learning platform is far from reality.

1.11.6 is Low-Code the Future of Software Development?

Low-code AI solutions are having a deep impact on the overall market right now, and its use is quickly expanding. For example, if you want to enhance your existing application, then either you can use a coding language such as Python or a low-code/no-code (LCNC) framework that has pre-designed and tested code blocks that can be instantly incorporated to use its functionality.

The forecast about the use of low-code/no-code development platforms follow a strongly positive curve. By the end of 2024, it is expected that more than 65% of applications will be developed using the low-code/no-code development approach. Also, more than 75% of large enterprises will be using at least four low-code/no-code development tools.

The demand for low-code/no-coding development platforms has increased tremendously and the worldwide low-code development technologies market is estimated to be $65 billion market cap by 2026. Businesses will benefit from low-code/no-code AI platforms in more data-driven sectors, such as marketing, sales, and finance. AI can help in predicting churn rates, analyzing reports, adding smart suggestions, automating invoicing and for lot more applications.

By 2030, the low-code development platform is expected to generate a revenue of $187 billion. As compared to no-code tools, low-code platforms provide a better scope of customization as per business requirements.

Chatgpt (Chat Generating Pre-Training)

ChatGPT is a language-generation software that can easily converse with people. For this, it answers follow-up questions, rejects inappropriate queries and even admits its mistakes (Source: OpenAI summary of the language model). To perform its task, ChatGPT has been extensively trained on an enormous amount of text data.

Using NLP techniques, training data and crawling the web to study archived books and Wikipedia, ChatGPT can recognize patterns to create text that mimics various writing styles. Some features of this amazing tool are as follows.

1. Facilitates communication in conversational format.
2. Uses Reinforcement learning techniques to respond more accurately and dynamically.

ChatGPT is developed by OpenAI using GPT-3.5 language.

We know that reinforcement learning learns by rewards and punishments received. In case of ChatGPT, a question is asked from the software. Then, a number of answers are sampled which are later manually ranked by humans. In fact, these ranks serve as training data for the reward model. The output of the reward model is then improved by further training a fine-tuned language model using reinforcement learning to react to queries. Therefore, human feedback plays a crucial role in the success of deploying this software.

Once trained, ChatGPT can respond to follow-up inquiries, acknowledge its errors, refute false assumptions and refuse unsuitable proposals.

Besides conversing, ChatGPT can respond to all types of writing, including theoretical essays, mathematical solutions and stories.

The underlying technique of ChatGPT can also alter how people use search engines by delivering answers to complex problems.

ChatGPT is a good debugging tool and can also fix the bug when you are asking a question.

ChatGPT is still in the research review stage. But still users can sign up and test it out for free.

Did you know that as soon as Elon Musk learned that OpenAI was using Twitter's database to train ChatGPT, he immediately stopped it from doing so because OpenAI is no longer open-sourced and non-profit, and it should eventually pay for this knowledge.

Chat GPT-3 be used to create more complex chatbots. Those chatbots can provide detailed information or perform difficult tasks like book flights, order food, automate simple tasks (like scheduling meetings, tracking appointments), notify about upcoming events, find local businesses, give personalized recommendations and advice based on preferences. Imagine a fitness freak person getting healthy recipes and exercise routines for a chatbot.

Big Concern: Human Replacement

Most significantly, ChatGPT has demonstrated the ability to build complex Python code and compose college-level essays. This has raised concerns that such technologies may eventually replace human workers like journalists or programmers.

Limitations

1. Chatbot is not a new thing. Several companies including Microsoft have developed them but did not get much success. We have read that in less than 24 hours, Twitter users taught Microsoft's Tay bot misogynistic and racist language. Meta released BlenderBot 3 which also disseminated racial, antisemitic and misleading information, including the assertion that Donald Trump won the 2020 presidential elections.

2. To prevent these kinds of incidents, OpenAI used a Moderation API (an AI-based moderation system) that notifies when language violates the company's content policy to stay away from communicating harmful or unlawful content. However, its moderation still has issues and is not perfect.
3. Like any other AI tool, ChatGPT is only as good as the data it is trained on. If the training data contains biases or inaccuracies, the model may reproduce these biases in its outputs.
4. ChatGPT sometimes write nonsensical answers and fixing this issue is challenging because there may be no source of truth and building more restrictions may result in declining questions that it can answer correctly.
5. ChatGPT gives different results when the questions are either slightly rephrased or asked multiple times.
6. Text generated by ChatGPT excessively uses certain phrases especially when trainers want longer answers that look more comprehensive.
7. Ideally, ChatGPT should ask clarifying questions when the user asks ambiguous queries but instead, it gives answers by guessing what the user intended.
8. Attempts were made to make ChatGPT refuse inappropriate requests, but it sometimes respond to harmful instructions or exhibit biased behaviour.
9. ChatGPT writes long paragraphs, poems, stories, etc. but it has not mastered the art of creative style, so its output has not yet topped best-seller lists. However, the program does write good bedtime stories for kids. But models like ChatGPT can never replace children's books that have been traditionally published.

Conclusion: Chat GPT is a powerful and versatile NLP tool that has the potential to revolutionize the way humans interact with machines.

Wishing you and everyone in your family a very healthy, wealthy, happy and prosperous 2023. Stay abundantly blessed always.

PROGRAMMER'S ZONE

The AI Game – Mystery Animal

Mystery Animal Game is a very interesting game in which the computer pretends to be an animal and the user tries to guess it by using his/her voice.

Rule 1: Users can only ask Yes-No questions such as 'Are you a reptile?' or 'Do you have feathers?' e 'Can you swim?', 'Are you a carnivore?', 'Are you a mammal'

Rule 2: Users have to guess the animal in 20 questions.

Visit the link to play the game. **https://mysteryanimal.withgoogle.com/**



Credit: Screenshot of STORY SPEAKING IN A GOOGLE DEVICE, **https://experiments.withgoogle.com/story-speaker**,is a trademark of Google LLC

Al Lab Session - Talk to Transformer

AI text generator tool, **Talk to Transformer** is used to create human-like text by predicting the next word from the 40 GB internet data (around 8 million web pages). The tool extensively uses Neural Network and NLP techniques (more specifically NLG) by learning from the data and by predicting the context of the previous data. Users just have to type a single line of text and the tool will in turn generate multiple results based on the prediction of what is already written.

Just visit the URL, **https://app.inferkit.com/demo**. Select a topic by clicking the down arrow and see the multiple results you get. Just read the sto-

ry. You will feel as if it had been written by humans only.

Fun Activity – Story Speaking in a Google Device

Use the following link to go to story-speaker page. **https://experiments.withgoogle.com/story-speaker**

Story Speaker lets anyone create talking, interactive stories with no coding required. The story can then be played anytime.



Credit: Screenshot of STORY SPEAKING IN A GOOGLE DEVICE, **https://experiments.withgoogle.com/story-speaker**,is a trademark of Google LLC

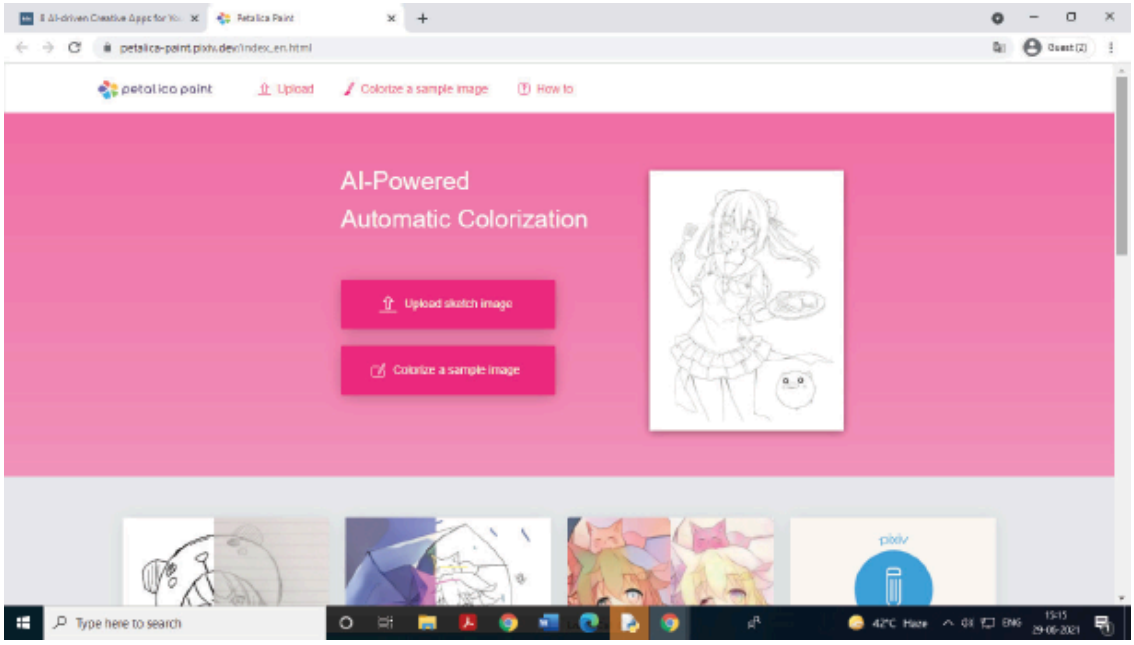
AI Lab Session for Painting Images

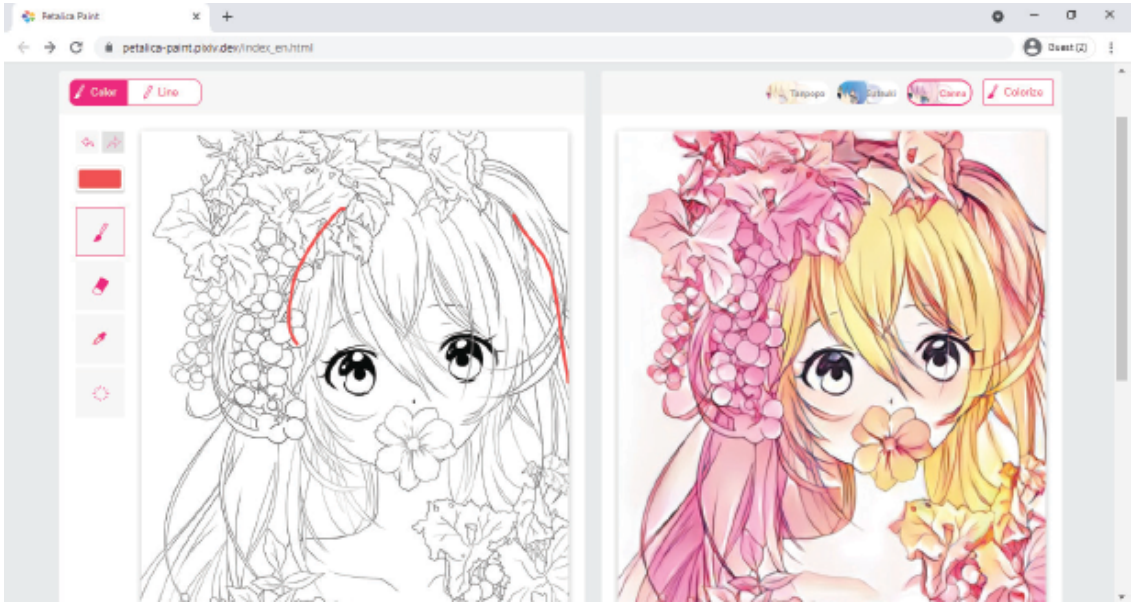
Visit the URL, **https://mitmedialab.github.io/GAN-play/**.

The pix2pix application converts images from one style to another using a machine learning model that is specifically trained on pairs of images. For example, you need to input a sketch/ drawing with outlines (or edges), select a model and then translate to see how the resultant image looks when filled with the selected pattern.

AI Lab Session – Colorizing Sketches

Visit the URL, **https://petalica-paint.pixiv.dev/index_en.html** You will be amazed to see that with Artificial Intelligence, you can draw within the tool or upload your sketches without worrying about the quality of the original file and the color them automatically in different ways.





AI Lab Session – Generate New Images

Visit the URL, **https://www.artbreeder.com/** The AI application allows users to keep selecting the most interesting image to discover totally new images. Therefore, Artbreeder turns the simple act of exploration into creativity by helping users make images by mixing any Artbreeder images together. Users can use any shared image for editing and mixing.



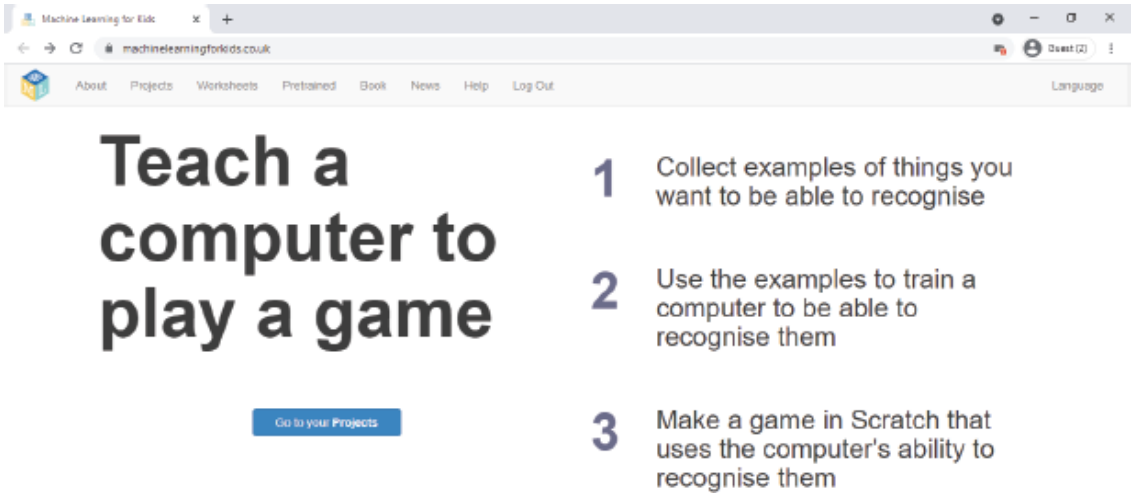
Credit: Joel Simon **https://www.artbreeder.com/**

AI Lab Session – Creating Comics from Video

Visit the URL **https://comixify.ai/?last=month&sortBy=trend** Upload a video and convert it into a comic. You can use this website to tell your story by making videos (less than 25 minutes) and then convert it into interesting story using comics text.

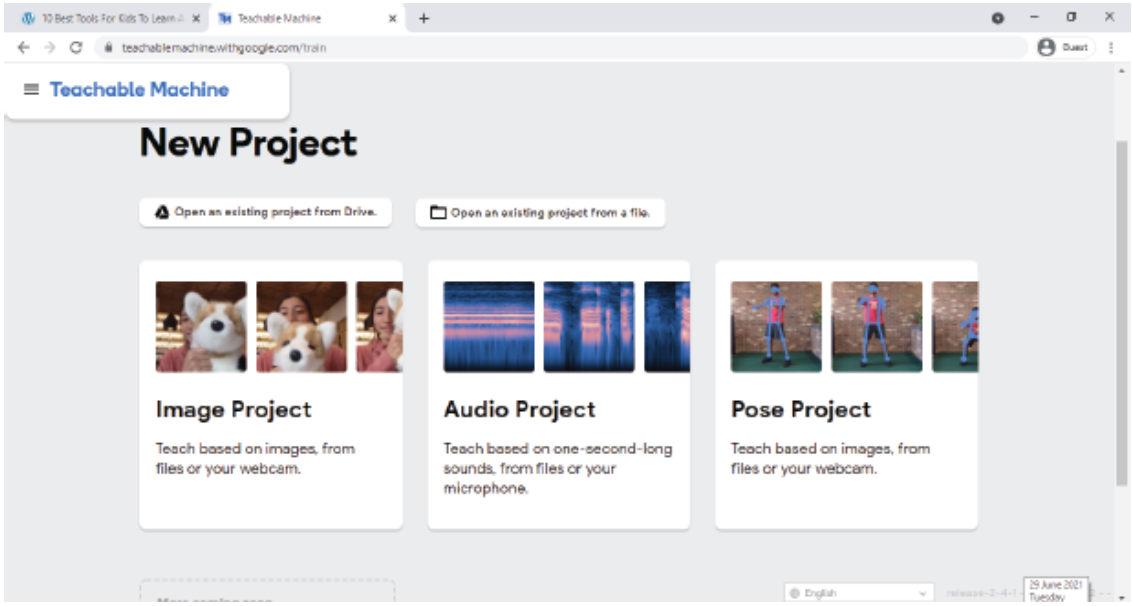
AI Lab Session – Train a Machine Learning Model

Visit the link **https://machinelearningforkids.co.uk/** and create a project that can classify text, images and numbers. You can train the model and test how accurate the results are.



AI Lab Session – Train a Machine Learning Model

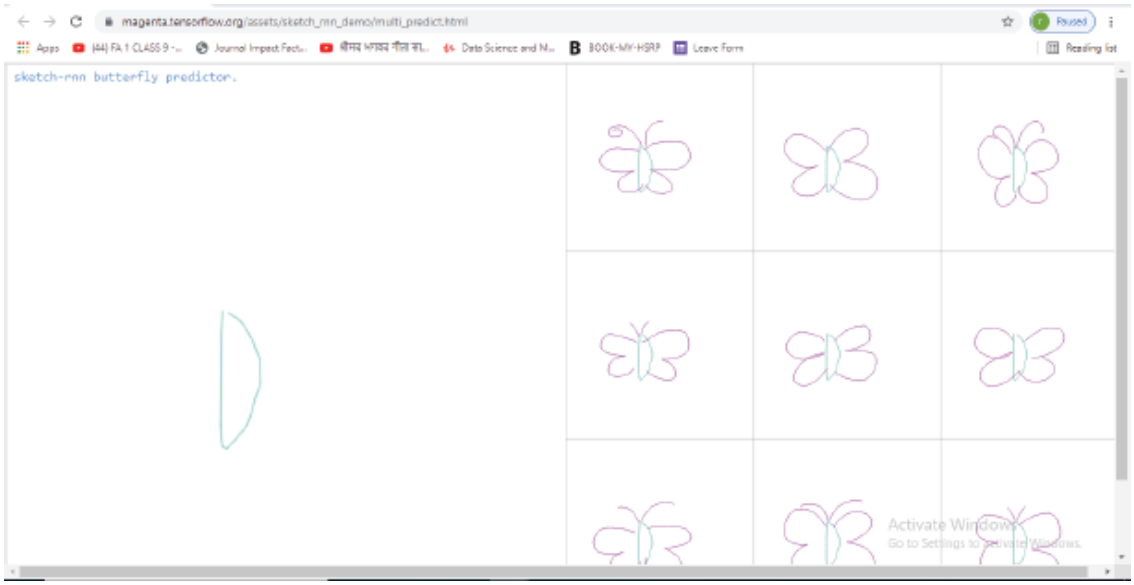
Visit the link **https://teachablemachine.withgoogle.com/** and create a project that can classify images, poses and sounds. You can train the model and test how accurate the results are.



Credit: Screenshot of TRAIN A MACHINE LEARNING MODEL, **https://teachablemachine.withgoogle.com/**,is a trademark of Google LLC

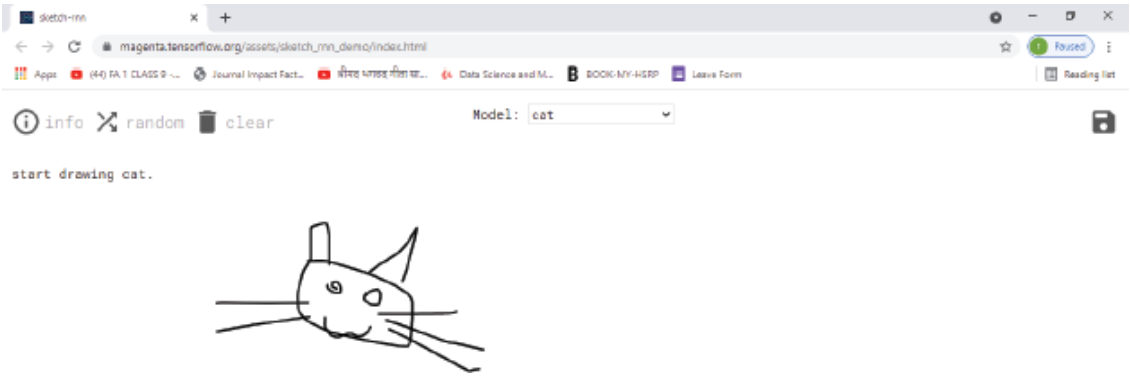
The experiment predicts the drawing which the user wants to create based on partial drawing made by the user. The user draws a sketch on the left side of the screen. On the right side, AI is used to predict the complete drawing based on user's inputs.

https://magenta.tensorflow.org/assets/sketch_rnn_demo/multi_predict.html



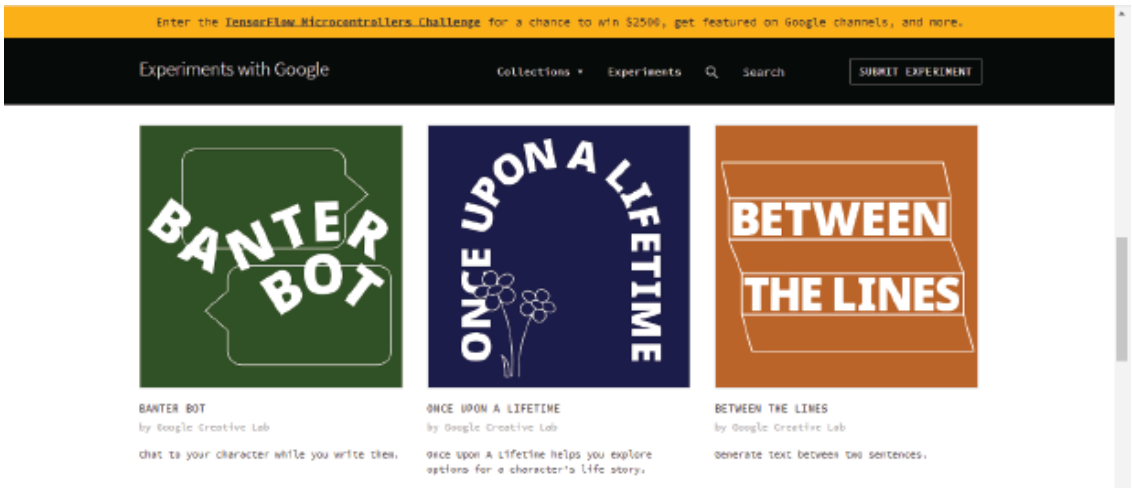
The experiment takes the drawing made by user as input and uses this input to make similar drawings. Visit the link **https://magenta.tensorflow.org/assets/sketch_rnn_demo/index.html** to draw interesting sketches. For convenience, users can select a category of the sketch before

drawing. However, do not forget to select a category and then draw entirely different sketch. For example, select cat in the category and draw a butterfly to see how the AI system behaves.



AI Lab Session – Experiments with Google

Google uses ML techniques to support authors create interesting stories with the help of three main tools that can be accessed by clicking on the URL **https://experiments.withgoogle.com/collection/aiwriting**



Credit: Screenshot of EXPERIMENTS WITH GOOGLE, **https://experiments.withgoogle.com/collection/aiwriting**, is a trademark of Google LLC

Machine learning models detect meaningful patterns in huge quantities of complex data. ML models can learn from larger patterns of grammar and semantics and use this learning on completely different contexts. The three tools that extensively assist users in creative writing include:

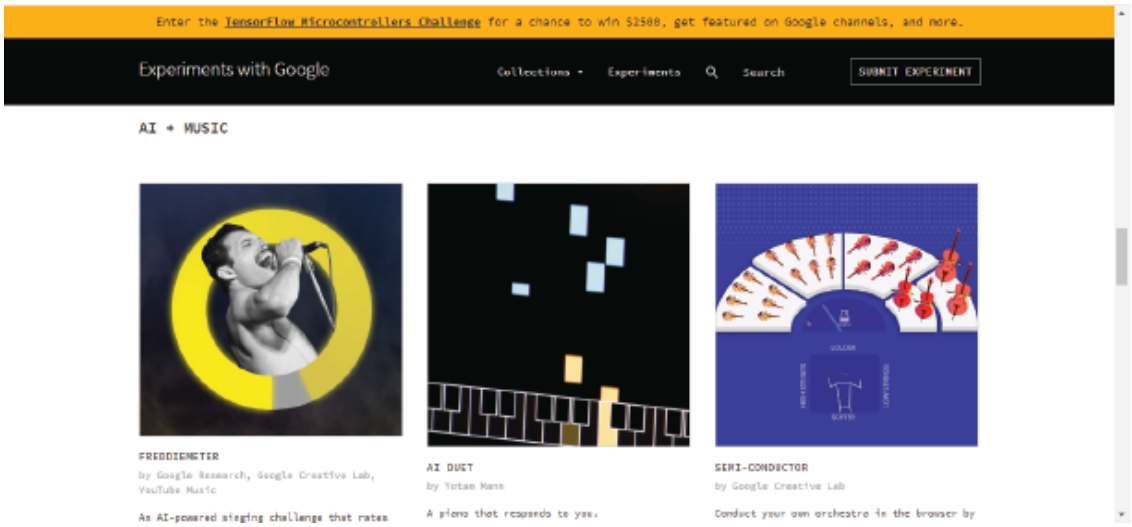
Between The Lines — The program takes the first and last line of a story and then uses machine learning techniques to interpolate between the lines and generate an interesting story specifying what would happen in the middle.

Once Upon A Lifetime— This is a character life story generator in which the user input keywords describing the kind of life they want to generate. This is especially important when writing biography of a character. The program creates the entire life story using the specified keywords.

Banter Bot— Banter bot is a character chatbot. Users can input information about what a character is like, and then can converse with it through text. As the user converses, the character evolves by learning from it.

AI Lab Session – AI for Music By Google

Visit the link to know about AI applications developed by Google to support music. **https://experiments.withgoogle.com/collection/ai**



Credit: Screenshot of AI FOR MUSIC BY GOOGLE, **https://experiments.withgoogle.com/collection/aiwriting**, is a trademark of Google LLC

FreddieMeter: It is an AI-powered singing challenge that rates how closely your singing matches the voice of Freddie Mercury. This application uses machine learning models to compare user's timbre, pitch, and melody with that of Freddie's. since user's audio is not uploaded to servers for analysis, so the vocal information stays private.

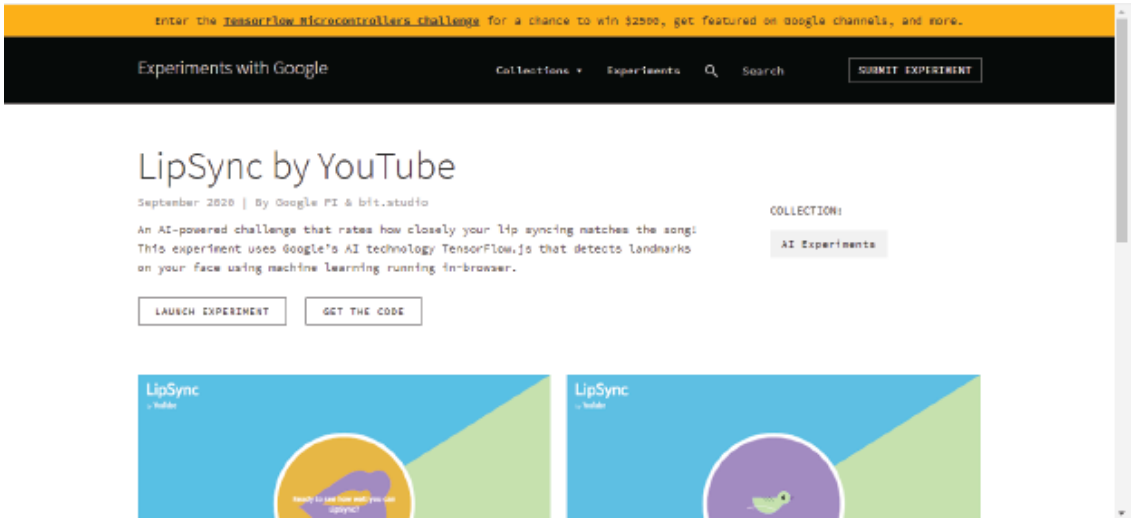
Semi-Conductor: This experiment by Google allows users to conduct their own orchestra in the browser just by moving their arms. Movement of the arms creates variation in the tempo, volume, and instrumentation of a piece of music.

AI Duet: It is another AI application by Google that plays a piano based on the inputs given by the user. Users can play some notes, and the computer will respond to the melody. Users may not even know how to play piano. They just have to press some keys and listen to what comes back.



AI Lab Session – Lip Sync

Go to the URL **https://experiments.withgoogle.com/lipsync** to launch the lip sync experiment developed by Google. The application uses AI techniques to user's lip syncing matches the song!



Credit: Screenshot of LIP SYNC, **https://experiments.withgoogle.com/collection/aiwriting**, is a trademark of Google LLC

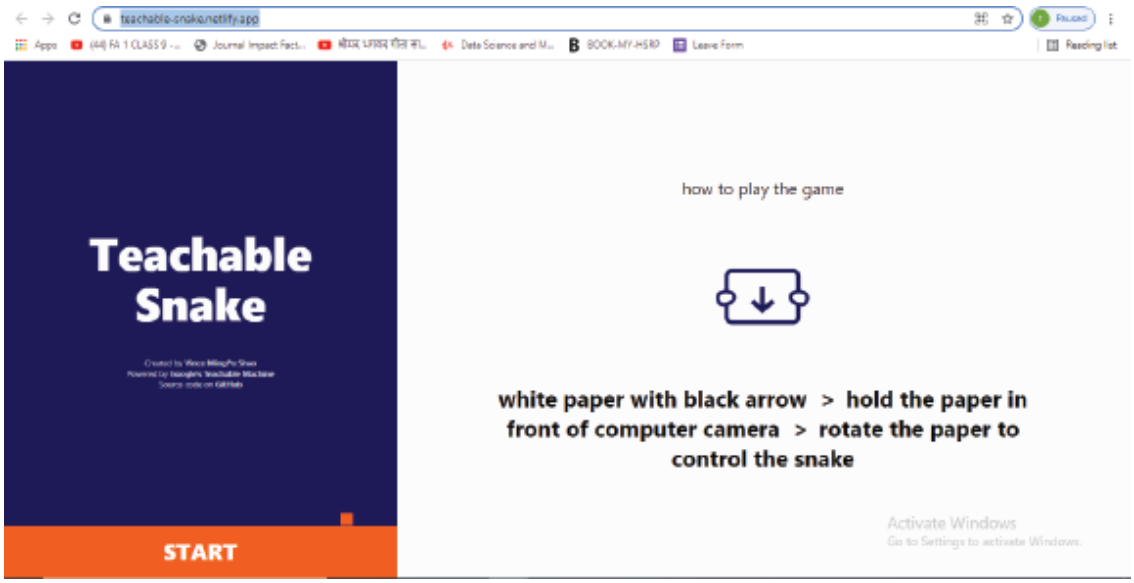
AI Lab Session – Touch Type By Google

Touch Type is an interactive type projection application that tracks simple body movements to replicate the effect on screen. Visit the link to **https://pose.yee.gd/** to use the application.

AI Lab Session – Teachable Snake

Teachable Snake is an interactive web game in which the users have to draw a black arrow on a piece of white paper as controller, and move the snake by turning the paper in different directions in front of webcam.

Visit the URL **https://teachable-snake.netlify.app/** to start the game.



Credit: Screenshot of teachable snake

AI Lab Session – Nsynth Sound Maker

The AI powered application help users create unusual new sounds using machine learning techniques. The application trained a neural network on over 300,000 instrument sounds. NSynth combine sounds, like a bass and flute, into a new, hybrid bass-flute sound. Visit the link

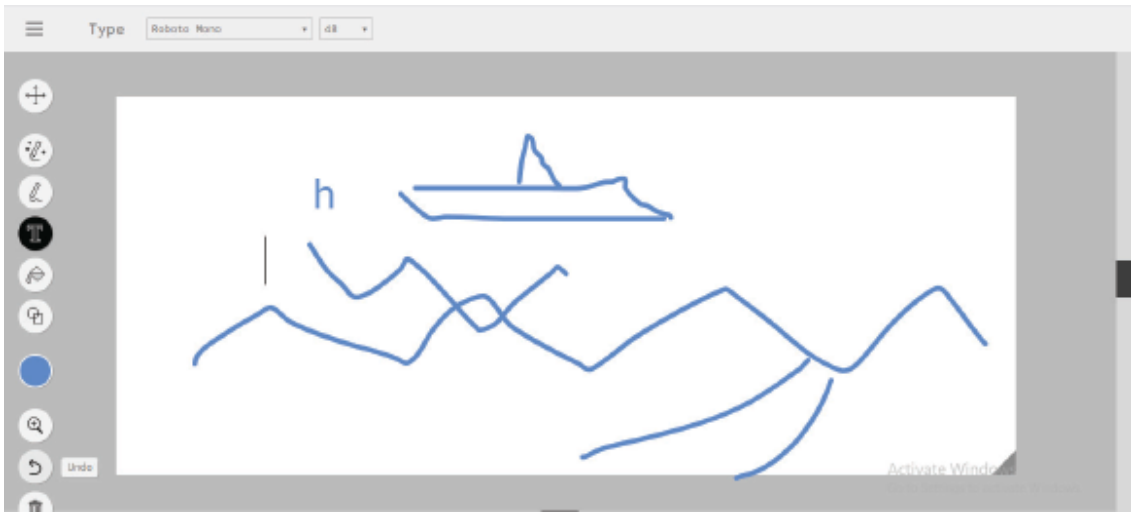
https://experiments.withgoogle.com/sound-maker to explore work with NSynth.



Credit: Screenshot of Nsynth SOUND MAKER, **https://experiments.withgoogle.com/sound-maker**, is a trademark of Google LLC

AI Lab Session - Autodraw

AutoDraw is an AI powered, fast drawing application for everyone. It is a new kind of drawing tool that uses ML techniques along with drawings from talented artists to help everyone create anything visual, fast. Users can use this application on their smartphone, tablet, laptop, desktop, etc free of cost without uploading/ downloading any data from the internet. Just visit the link **https://www.autodraw.com/** to use this application.



AI Lab Session - Scroobly

Scroobly is a Google experiment that is used to create fun animations in real-time with your camera. The application uses machine learning (ML) models to map user's live motion to the screen doodle. This means that the ML system updates the animation seen on screen according to user's movement. Using Scroobly, users are extensively use artificial intelligence as a creative tool to become a digital animator. All this can be easily done

without any prior knowledge of computer coding. Visit the link **https://www.scroobly.com/** to experiment with Scroobly.

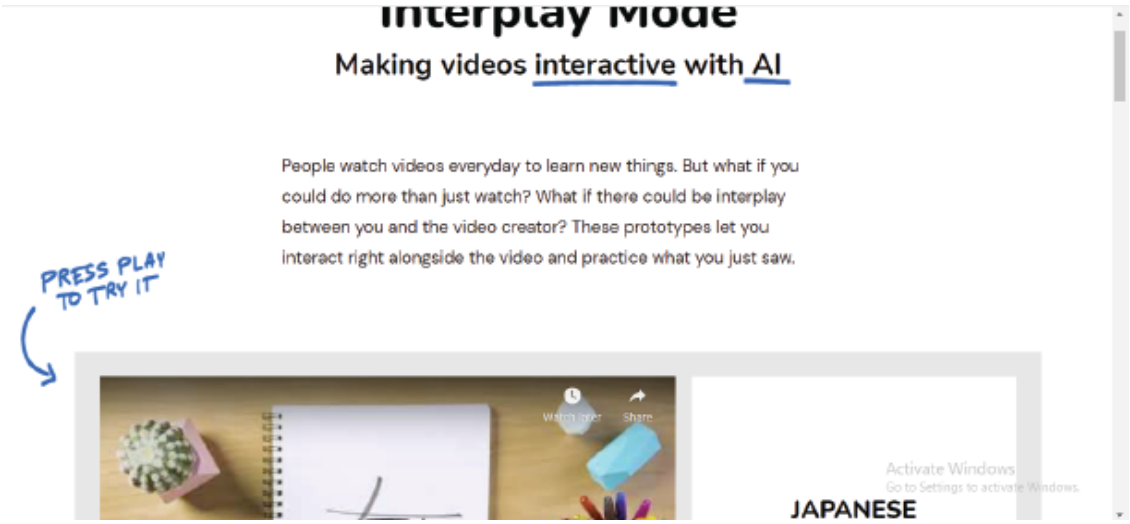


Credit: Screenshot of SCROOBLY, **https://www.scroobly.com/**, is a trademark of Google LLC

AI Lab Session – Interplay Mode

Interplay Mode is a Google experiment that allows users to make videos interactive with AI. With online education, students often learn something new from videos. But videos have a limitation of being a one-way communication mode. Therefore, the Interplay mode is extremely useful as it enables users to right alongside the video and practice what you have just learnt. So, now there is an interplay between the user and the video creator, to facilitate interaction. Visit the given link to launch the Interplay Mode application by Google.

https://experiments.withgoogle.com/interplay-mode/view/

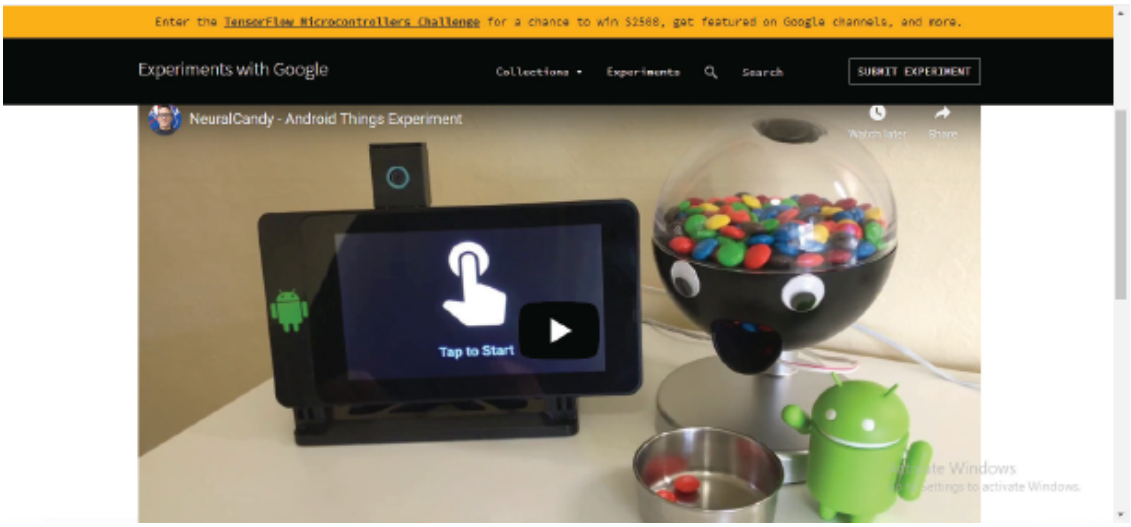


Credit: Screenshot of INTERPLAY MODE, **https://experiments.withgoogle.com/interplay-mode/view/**, is a trademark of Google LLC

AI Lab Session - Neuralcandy

NeuralCandy is an interesting experiment in which a machine dispenses candy when the user displays the requested image in front of the trained image model. The application prompts the user for a random image to be placed in front of the camera module. If the image matches the request; then the motor of the candy dispenser is activated to release the reward.

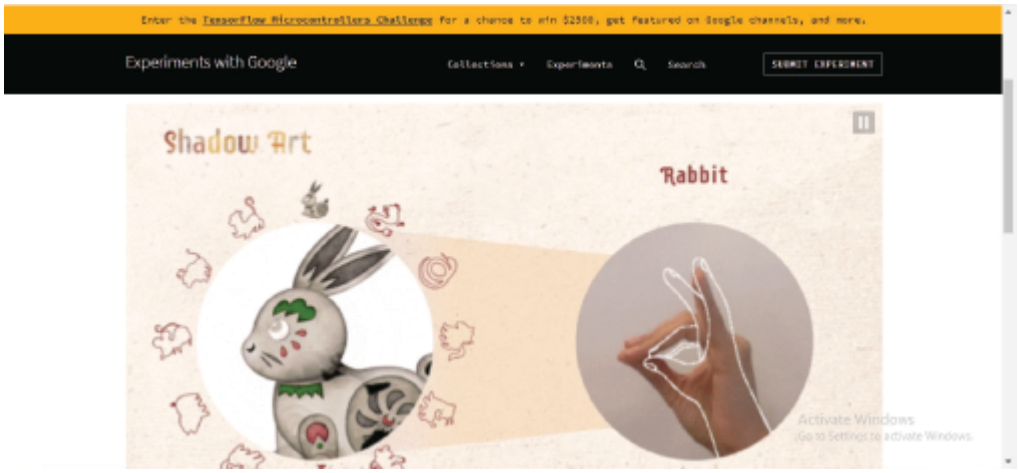
You can use a smartphone to search on Google for the requested target image and put it in front of the Pi camera. The game can be started by visiting the URL **https://experiments.withgoogle.com/neuralcandy**



Credit: Screenshot of NEURALCANDY,**https://experiments.withgoogle.com/neuralcandy**, is a trademark of Google LLC

AI Lab Session – Shadow Art

Shadow Art **i**s an application by Google that allows users to use AI to play with the ancient art of shadow puppetry. To play with this game, users have to create shadow puppets with hands in front of laptop or phone camera. The puppets correspond to one of the lunar cycle's 12 zodiac symbols. The ML model identifies the animal formed by user's hands. If the user is accurately able to form the first puppet within the specified time then the next zodiac symbol has to be tried until all 12 puppets are correctly formed by the user. Visit the link to launch the Google experiment **https://experiments.withgoogle.com/shadow-art**.



Credit: Screenshot of SHADOW ART, **https://experiments.withgoogle.com/shadow-art**., is a trademark of Google LLC

AI Lab Session – Imaginary Soundscape

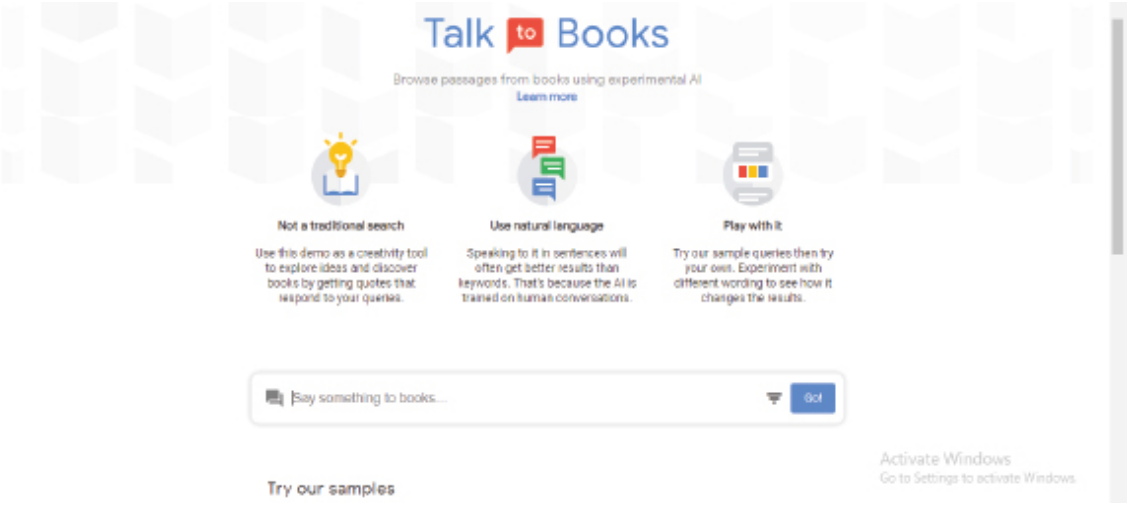
Imaginary Soundscape is a Web-based sound installation on Google Street view. Created using deep learning models, the application facilitates users to imagine sounds by taking a glance at a photo. For example, a beach scene is displayed with the sound of crashing waves. Similarly, when viewing a busy road, the sounds of horns and street advertising can be heard. 'Imaginary Soundscape' allows users to freely walk around Google Street View and immerse themselves into imaginary soundscapes generated using deep learning techniques. To use this application, visit the URL **http://www.imaginarysoundscape.net/**



Credit: Qosmo, © 2023 Qosmo, Inc

AI Lab Session – Talk to Books

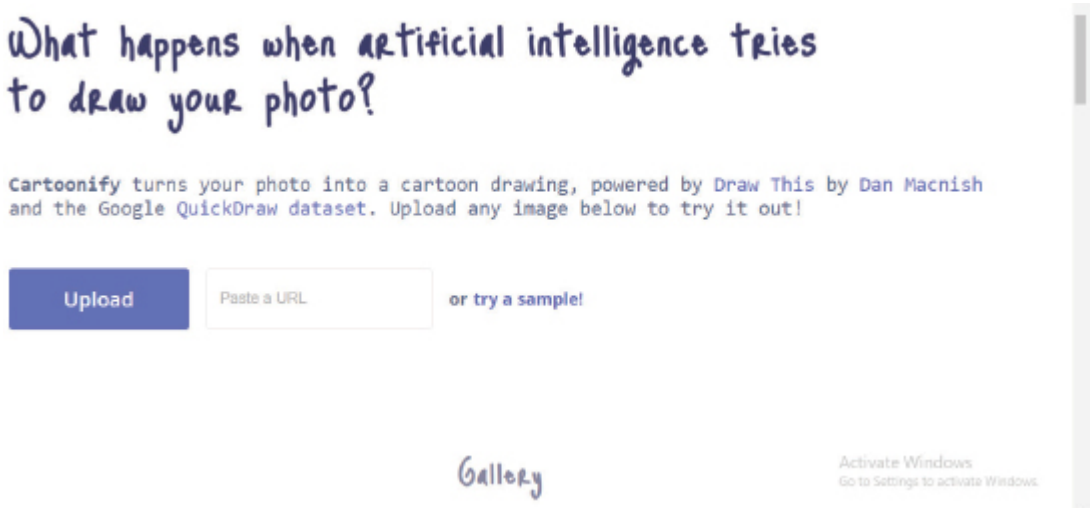
Talk to Books is an AI powered application that allows users to browse passages from books thereby providing an absolutely new way to interact with books. Users can speak directly with all 100,000 books, ask questions or make statements. Users can even filter their search results depending on category of books. Visit the link to use this application **https://experiments.withgoogle.com/talk-to-books**



Credit: Screenshot of TALK TO BOOKS, **https://experiments.withgoogle.com/talk-to-books**, is a trademark of Google LLC

AI Lab Session - Cartoonify

Cartoonify **i**s an experiment by Google that use Artificial Intelligence to turn a photo into a cartoon using neural networks. For best results, upload images with clear objects on a solid background. So visit the link **https://www.kapwing.com/cartoonify** to cartoonify yourself !!!



Credit: Cartoonify, © 2023 Kapwing

AI Lab Session - Fontjoy

Fontjoy is an AI powered Google experiment that allows users to generate font combinations using deep learning techniques. Visit the link **https://fontjoy.com/** and click on Generate button to create a new font pairing. Click on the Lock icon to lock fonts that you want to keep, and click on Edit to choose a font manually. The text is editable, so users can anytime replace the default text with their own piece of text.

The application is mainly used to select fonts that share an overarching theme yet have a pleasing contrast. With this application, we are using neural networks to visualize Which fonts work together.

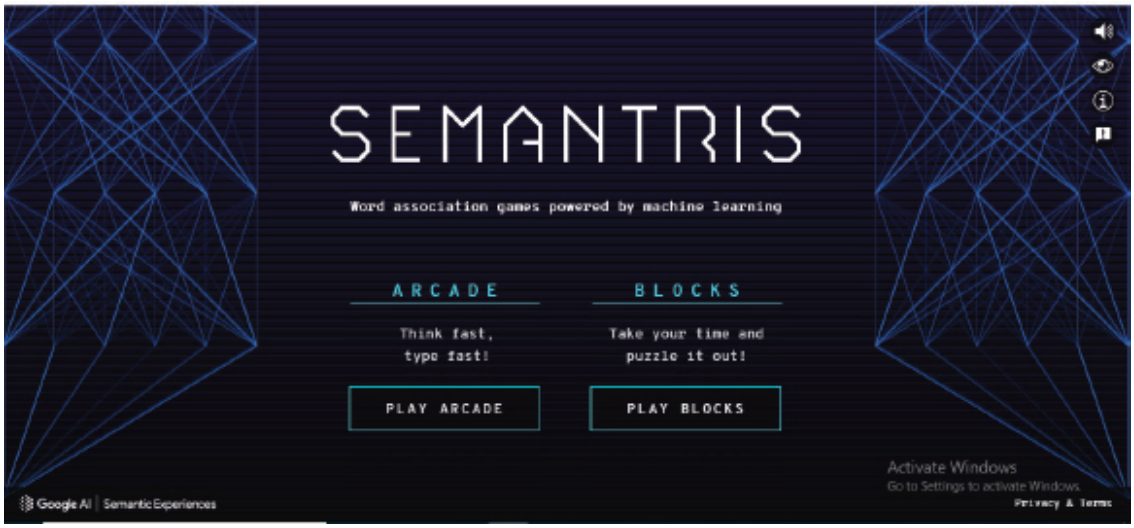
Classifying Images Activity

Visit the link **https://www.cs.ryerson.ca/~aharley/vis/conv/flat.html** to explore an online application of classifying different numbers. You can visualize how analysis is done using multiple convolution layers to finally classify the input digit.

AI Lab Session - Semnatris

Semantris is a word association games powered by semantic search. The application uses machine-learning and more specifically NLP techniques. Each time the user enters a clue, the AI looks at all the words in play and chooses the ones it thinks are most related. To develop this game, the AI system was trained on billions of examples of conversational text that

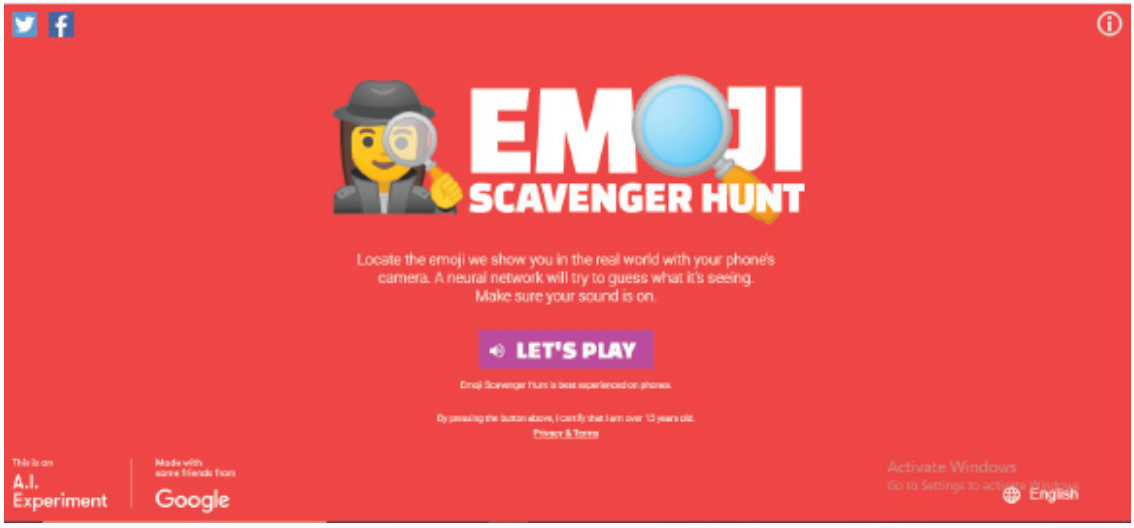
span a large variety of topics. So, the application can make many types of accurate associations. Visit the link **https://research.google.com/semantris/** to launch this experiment.



Credit: Screenshot of EMNATRIS, **https://research.google.com/semantris/**, is a trademark of Google LLC

AI Lab Session – Emoji Scavenger Hunt

Emoji Scavenger Hunt is an AI powered Google experiment that identifies emojis in the real world with user's device's camera. The application uses neural networks to identify the real-world versions of the emojis we use every day. To launch this application, visit the link **https://emojiscavengerhunt.withgoogle.com/**



Credit: Screenshot of EMOJI SCAVENGER HUNT,**https://emojiscavengerhunt.withgoogle.com/**, is a trademark of Google LLC



KEY TERMS

Artificial Intelligence (AI): The science and engineering of making intelligent machines, especially intelligent computer programs.

Learning processes: Acquiring data and creating rules for turning that data into actionable information.

Reactive machines: Basic machines with no memory to store and thus use past experiences to determine future actions.

Limited memory machines: Machines that retain data but only for a short period of time and can use it only for a limited time. They cannot keep adding data permanently to a library of their experiences.

Robotics: A branch of engineering that involves the conception, design, manufacture and operation of robots.

Drone: Any unpiloted aircraft that operates using a combination of technologies including computer vision, artificial intelligence, object avoidance tech and others.



RECAPITULATION POINTS

- AI systems work effectively when fed with a large amount of labelled training data.
- AI programming focuses on three cognitive skills: learning, reasoning and self-correction.
- Long Short-Term Memory (LSTM) models use past data to help predict the next item in a sequence. More recent information is given a higher priority for prediction than past data.
- Augmenting robots either enhance current human capabilities or replace the capabilities a human may have lost. Robotic prosthetic limbs or exoskeletons used to lift hefty weights are examples of such robots.

- Robotic airplanes can fly without a pilot aboard. NASA is also developing Robonaut to help people in space. The upper body of Robonaut looks like a person. It has a chest, head and arms.
- AI and machine learning can be used to identify and predict potential threats and catch instances of fraud especially those involving finances.
- No-code tools allow people to build applications and systems without having to write long and complex program codes. Applications can now be designed using visual interfaces and guided user actions.
- Low-code development platforms are based on the concepts of model-driven design, automatic code generation and visual programming.



ANSWER THE QUESTIONS

1. Define AI and give at least three advantages and three disadvantages of this technology.
2. How do AI systems work?
3. What are the three cognitive skills on which AI focuses?
4. Differentiate between the following:
    1. Weak AI and strong AI
    2. Reactive machines and limited memory AI systems
    3. Augmented and artificial intelligence
    4. No-Code and low-code AI
    5. Traditional tools and low-code AI tools
5. How will you categorize AI systems based on their functionality?
6. Explain machine learning models that apply limited memory artificial intelligence.
7. Give any ten applications of AI.
8. What are the different types of robots?
9. Give any five uses of robots.
10. How are drones useful in the field of military, agriculture and personal transportation?
11. Give some pros and cons of no-code and low-code AI tools.
12. What is the reason behind tremendous growth of no-code AI tools?

FILL UP

1. In _____, _____ coined the term artificial intelligence.
2. _____ AI is specifically designed to perform a specific type of task.
3. _____ focuses on imitating the human brain.
4. _____ machines interpret user's feelings by learning not only what they communicate to them but also how they communicate it.
5. Sophia and Atlas are two popular examples of _____robots.
6. _____ allows machines to obtain information from digital images, videos, pdfs and other visual data, and take actions based on their learnings.
7. _____ is a free desktop application by Microsoft that is used for image classification with object detection.
8. _____ is a low-code AI tool that quickly creates a custom AI vision solution.

STATE TRUE OR FALSE

1. AI systems work well with labelled data.
2. Siri and Alexa are weak AI systems.
3. Weak AI makes full attempt to resemble human brain.
4. Autonomous vehicles use reactive machine systems.
5. Long short-term memory (LSTM) models do not consider past data while making predictions.
6. Pre-programmed robots operate in a controlled environment to perform simple, monotonous tasks.
7. Drones are being used to provide Internet access in remote areas and over irregular landforms.
8. Predicting churn rate is an application of computer vision.



MULTIPLE CHOICE QUESTIONS

1. Select a language that is not used to code AI applications.
    1. R
    2. Python
    3. Java
    4. C++
2. The CEO of which company said ‘AI First’?
    1. Microsoft
    2. Google
    3. Amazon
    4. Twitter
3. Choosing the right algorithm to reach the desired outcome is a part of which process?
    1. Learning
    2. Reasoning
    3. Self-correction
    4. AI
4. In 1958, John McCarthy developed the AI programming language _____.
    1. Lisp
    2. Prolog
    3. R
    4. Python
5. Which company made the first self-driving car that could pass a state driving test?
    1. Microsoft
    2. Google
    3. Amazon
    4. Twitter
6. Google search, image recognition software, self-driving cars and IBM’s Watson are some examples of which systems?
    1. Weak
    2. Strong
    3. Both of these

4. None of these

7. IBM’s Deep Blue is an example of _____
    1. Limited memory
    2. Reactive machines
    3. Theory of mind
    4. Self-awareness
8. Which machine makes decisions keeping in view the feelings experienced through self-reflection and determination?
    1. Reactive
    2. Limited memory
    3. Theory of mind
    4. Self-awareness
9. _____ robots are designed to perform tasks in open environments that do not require human supervision.
    1. Pre-programmed
    2. Humanoid
    3. Augmenting
    4. Autonomous
10. Drones are examples of _____ robots.
    1. Pre-programmed
    2. Tele-operated
    3. Augmenting
    4. Autonomous



FIND OUT THREE AI TOOLS FOR

Content writing

Text paraphrasing

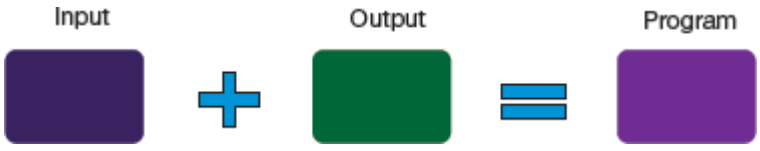
Note making during a meeting

Low code AI tools

No code AI tools

APPLICATION BASED QUESTIONS

1. List two real-world applications of supervised and unsupervised learning.
2. Imagine that you are supposed to train a machine to play chess. What kind of learning algorithm do you think would work better?
3. Your Instagram posts are unstructured. Justify this statement.
4. What parameters will you use to train the machine learning model to identify a cycle, auto and a car?
5. Given an image having both dogs and cats which the machine has not seen before, how will the ML model categorize them if no training data or examples were provided for prior training?
6. If you meet a stranger at a public place, how will you classify this person—gentle man/woman or a fraudster?
7. In a cricket match, mention any two parameters that you will use to classify players?
8. Can you think of any three job profiles that existed in the 80s but no longer relevant now?
9. Can you think of any three job profiles that did not exist in the 80s but are very demanding these days?
10. Can you think of any three job profiles that do not exist now but may become popular in next 10 years?
11. Think of any two problems that you would like to solve using machine learning.
12. Imagine you have been assigned the task to design a recommendation system for the school library. What data do you need to collect for training the recommendation system?
13. Fill in the boxes to predict customers who will pay the bills late.



LINKS TO EXPLORE

Go through these links to learn more about IBM Watson, Google Driverless car and Sophia, the humanoid.

https://www.youtube.com/watch?v=s_wgf75GwCM

https://www.youtube.com/watch?v=cdgQpa1pUUE

https://www.youtubeom/watch?v=cdgQpa1pUUE

Visit these links to know more about Honda Asimo and Boston Dynamics AI Robot. Make a beautiful presentation to highlight their capabilities.

(**https://www.youtube.com/watch?v=1urL_X_vp7w**)

https://www.youtube.com/watch?v=NR32ULxbjYc

Open the URL Crime in India - 2021 | Open Government Data (OGD) Platform India (**https://data.gov.in/catalog/crime-india-2021**).and answer the following questions.

1. What kind of data does it hold?
2. Why does the Government of India publish this data?
3. Why is this data available on public domain?
4. What do you think, from where did the government get this data?

ANSWERS

Fill Up

1. 1956, John McCarthy
2. Weak/Narrow
3. Theory of mind
4. Self-awareness AI

5. humanoid
6. Computer vision
7. Lobe
8. viso.ai

State True or False

1. True
2. True
3. False
4. False
5. False
6. True
7. True
8. False

Multiple Choice Questions

1. d
2. b
3. c
4. a
5. b
6. a
7. b
8. c
9. d
10. b